MET 2026 + NIMHANS Prep, Full Notes

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About this book

These are the consolidated study notes built for the MET 2026 (23 May) and NIMHANS MA Clinical Psychology (24 May) entrance exams. Every chapter is written as a story you can read end-to-end, not a fragment list. The biopsychology chapters in particular were built after a diagnostic showed action potential / ion channels / synaptic transmission as the biggest leverage point.

Reading order suggestion: 1. Action Potential (read this first if biopsych mechanics feel shaky) 2. Sleep 3. Brain Anatomy 4. Neurons and Glia 5. Biopsychology Master Cheatsheet 6. Psychological Assessment Primer (story form) 7. Reliability and Validity (formulas) 8. Assessment Master Cheatsheet 9. Diagnostic Baseline (your own mistakes from 2026-04-22)

Part 1: Biopsychology

Action Potential — Full Primer

Action Potential — Complete MET Reference

The single most testable neurophysiology topic across MET, NIMHANS, AIIMS. Built around ion movement (Na+, K+, Ca2+, Cl-), the four phases, and the rules that make MCQs predictable.

0. What an Action Potential Actually Is (start here)

An action potential is the basic signal a neuron uses to talk. It is a short (~1 millisecond) electrical pulse that travels down the axon from the cell body to the axon terminal. When it reaches the terminal, it causes neurotransmitter release, which then signals the next neuron. Every thought, sensation, movement, and emotion you have is built out of millions of these pulses firing in patterns. So the question "how does a neuron fire?" = "how does an action potential happen?"

The one-sentence story

When a neuron is at rest, the inside of its membrane is **negative** relative to the outside (about -70 millivolts). When the neuron is stimulated enough, ion channels open in a precise sequence — first sodium (Na+) rushes IN making the inside briefly **positive**, then potassium (K+) rushes OUT bringing it back to **negative** — and that pulse of voltage change propagates down the axon at up to 120 m/s. That's it. Everything else in this primer is the mechanics of how that pulse is generated, shaped, and propagated.

Five things you must know before the mechanics make sense

1. **Neurons have a charged membrane at rest.** Inside is at -70 mV, outside is the reference (0 mV). This charge difference is called the **resting membrane potential (RMP)**. It exists because the membrane lets some ions cross more easily than others (mostly K+ leaks out a little, leaving the inside negative).

2. **The membrane is studded with ion channels.** Some are always open ("leak channels"), some open only when voltage changes ("voltage-gated"), some open only when a chemical binds ("ligand-gated"). The action potential is driven by **voltage-gated** channels for Na+ and K+.

3. **Ions flow because of two forces: concentration and charge.** Na+ wants to enter (lots outside, little inside, attracted to negative interior). K+ wants to leave (lots inside, little outside, but pulled back by negative interior). When a channel opens, the relevant ion moves whichever way its forces dictate. Knowing which ion goes which way at each phase is the whole game.

4. **An action potential is "all or none."** If the stimulus is strong enough to push the membrane to about -55 mV (the **threshold**), a full action potential fires. If the stimulus is weaker, nothing happens. Bigger stimuli do not produce bigger APs — they produce more APs per second (frequency coding). One AP looks identical to every other AP in that neuron.

5. **The signal travels because each segment of axon triggers the next segment.** Na+ entering at one spot depolarizes the next bit of membrane to threshold, which opens its Na+ channels, and so on. In myelinated axons, the AP "skips" between exposed gaps (Nodes of Ranvier), which is way faster than continuous conduction.

What an action potential looks like on a graph

Voltage on Y-axis (in mV), time on X-axis (in milliseconds). The whole thing takes about 2 ms.

```
+30 ─────────────●─── PEAK (Na+ channels inactivate)
                /|\
               / | \
              /  |  \ ← REPOLARIZATION (K+ flowing OUT)
             /   |   \
            /    |    \
−55 ────●─/──────|──────●───── THRESHOLD (−55 mV)
       /         |       \
      /    ↑     |        \___
−70 ─●   DEPOL   |            \____ ← AFTERHYPERPOLARIZATION
     |   (Na+ IN)|                  (K+ channels still open
RMP                                  briefly, overshoots RMP)
```

Why exams hammer this topic

Action potential MCQs test precise, factual recall of: - **Which ion moves at which phase** (Na+ in for depolarization, K+ out for repolarization, Ca2+ in at the terminal for NT release, Cl- in for IPSP) - **Which channels open and when** (voltage-gated Na+ at threshold, voltage-gated K+ slightly later, voltage-gated Ca2+ at the axon terminal) - **Refractory periods** (absolute = Na+ inactivated, relative = K+ still open) - **Conduction speed and myelin** (saltatory in myelinated fibers, fastest in A-alpha) - **Synaptic transmission cascade** (AP arrives → Ca2+ in → vesicles fuse → NT released → binds postsynaptic receptors → EPSP or IPSP)

If you know the story below cold, you will get every single one right. If you confuse Na+ and K+, mix up depolarization and repolarization, or don't know that Ca2+ triggers NT release, you will bleed marks. The 6 questions you got wrong on the diagnostic were all in this cluster.

Now that the big picture is set, here is the mechanics in detail.

1. Resting Membrane Potential (RMP)

The neuron at rest sits at about **-70 mV** (range -60 to -90 depending on neuron type). Inside is negative relative to outside.

Why -70 mV? Three forces.

Force	What it does	Effect
Concentration gradient	K+ wants to leave (high inside), Na+ wants to enter (high outside)	Sets up the chemistry
Electrical gradient	Inside is negative, attracts cations	Opposes K+ leaving
Selective permeability at rest	Membrane is ~50x more permeable to K+ than Na+ (leak channels)	K+ dominates

The membrane settles where K+ efflux balances the electrical pull keeping K+ in. That equilibrium = ~-70 mV (close to K+'s equilibrium potential of -90 mV, dragged up by small Na+ leak).

Ion concentrations (mEq/L), memorize this table

Ion	Inside	Outside	Equilibrium potential (Eion)
K+	140 (high)	5 (low)	-90 mV
Na+	15 (low)	145 (high)	+60 mV
Cl-	10 (low)	110 (high)	-65 mV
Ca2+	0.0001 (very low)	2 (high)	+120 mV

Memory hook: "K is in, Na is out, Cl is out, Ca is way out."

Na+/K+ pump (Na+/K+ ATPase)

- Pumps **3 Na+ out** for every **2 K+ in**, using 1 ATP.
- **Electrogenic**: net export of +1 charge per cycle, contributes ~-4 mV to RMP.
- Maintains the gradients but does NOT directly cause RMP. RMP is caused by K+ leak channels following the K+ gradient. The pump just keeps the gradient charged up.

Common MCQ trap: "What sets resting potential?" Answer: K+ permeability via leak channels. NOT the Na+/K+ pump (it's the gradient-maintainer, not the voltage-setter).

2. The Four Phases of an Action Potential

```
       +30 mV ─┐      ┌─ Peak (Na+ inactivation)
               │    │
               │   /│
               │  / │ ← REPOLARIZATION (K+ out)
               │ /  │
               │/   │
   Threshold ──┼────┼──────────── −55 mV
   ──────────/│    │\
            / │    │ \
   RMP ────/──┘    │  \____      ← AFTERHYPERPOLARIZATION (K+
still open)
   −70 mV         │      \
                  │       \____ back to RMP
                  └─ DEPOLARIZATION (Na+ in)

   Phase 1: Depolarization     → Na+ channels open, Na+ rushes IN
   Phase 2: Repolarization     → Na+ inactivates, K+ channels
open, K+ rushes OUT
   Phase 3: Hyperpolarization  → K+ channels still open, briefly
overshoots RMP
   Phase 4: Return to RMP      → K+ channels close, Na+/K+ pump
restores ions
```

Phase-by-phase ion movement

Phase	What opens	Ion movement	Voltage change
Threshold reached (-55 mV)	Voltage-gated Na+ channels	Na+ starts in	Trigger
1. Depolarization	Na+ channels fully open	Massive Na+ in	-55 → +30 mV
Peak (+30 mV)	Na+ channels inactivate	Na+ stops	Plateau briefly
2. Repolarization	Voltage-gated K+ channels open (slower)	Massive K+ out	+30 → -70 mV
3. Hyperpolarization (afterhyperpolarization)	K+ channels still open	More K+ out	-70 → -80 mV
4. Return to RMP	K+ channels close, pump works	Pump restores Na+/K+	-80 → -70 mV

Memory hook: "**Na+ in, K+ out**" — that's the whole story. Na+ rises = depolarization. K+ exits = repolarization.

3. The All-or-None Principle

- If stimulus reaches threshold (-55 mV typical), an AP fires at full amplitude.
- If stimulus is below threshold, **nothing happens**. No partial AP.
- Stronger stimulus = more frequent APs (rate coding), NOT bigger APs.

MCQ trap: “How does the brain encode stimulus intensity?” Answer: **frequency of action potentials**, not amplitude. Amplitude is fixed.

4. Refractory Periods

Two types, both critical MCQ topics.

Absolute refractory period

- Duration: about **1-2 ms**.
- Cause: **Na+ channels are inactivated** (not just closed; they have an inactivation gate).
- Effect: **No new AP can fire**, no matter how strong the stimulus.
- Function: ensures one-way propagation of AP down axon.

Relative refractory period

- Duration: about **2-4 ms** after absolute.
- Cause: **K+ channels still open** (membrane is hyperpolarized).
- Effect: A new AP CAN fire, but requires a **stronger-than-normal stimulus**.
- Function: limits maximum firing rate (~1000 Hz max).

MCQ trap: Inactivated ≠ closed. Na+ channel has 3 states: closed (resting), open (during AP), inactivated (after AP, “ball and chain” blocking pore). Resets only after repolarization.

5. Propagation Down the Axon

Continuous (unmyelinated)

- AP regenerates point-by-point along the membrane.
- Slow: **0.5 - 2 m/s**.
- Found in: C fibers (slow pain, temperature, postganglionic autonomic).

Saltatory (myelinated)

- AP “jumps” between **Nodes of Ranvier** (gaps in myelin).
- Voltage-gated Na+ channels concentrated only at nodes.
- Fast: **up to 120 m/s** in large myelinated A-alpha fibers.
- Found in: A fibers (motor, proprioception, fast pain, touch).

Memory hook: “Saltatory” = Latin *saltare*, “to leap.”

Factors that increase conduction velocity

1. **Myelination** (biggest factor).
2. **Axon diameter** (bigger = faster, less resistance).
3. **Temperature** (warmer = faster, up to a limit).

Demyelinating diseases (multiple sclerosis, Guillain-Barre): slow or block conduction → numbness, weakness, vision loss.

6. Synaptic Transmission (the post-AP step)

When the AP reaches the axon terminal:

1. **Ca2+ channels open** (voltage-gated). Ca2+ rushes IN.
2. **Ca2+ triggers vesicle fusion** with presynaptic membrane (SNARE proteins, synaptotagmin = Ca2+ sensor).
3. **Neurotransmitter released** into synaptic cleft.
4. **Binds postsynaptic receptors** → ion flow → EPSP or IPSP.
5. **Cleared** by reuptake, enzymatic breakdown, or diffusion.

Why Ca2+? It’s the trigger ion at the axon terminal. Without Ca2+ entry, no transmitter release.

EPSP vs IPSP

Type	Ion movement	Effect on postsynaptic neuron
EPSP (excitatory post-synaptic potential)	Na+ in (or Ca2+ in)	Depolarizes, brings closer to threshold
IPSP (inhibitory post-synaptic potential)	Cl- in (or K+ out)	Hyperpolarizes, moves away from threshold

Summation rules: - **Spatial:** multiple synapses fire at the same time, signals add. - **Temporal:** one synapse fires repeatedly, signals add over time. - AP fires only if summed depolarization reaches threshold at the **axon hillock** (the AP initiation site).

7. The Top 10 MCQ Traps on Action Potentials

1. **What sets RMP?** K+ permeability (leak channels), not the Na+/K+ pump.
2. **What ion triggers neurotransmitter release?** Ca2+ at the axon terminal.
3. **All-or-none refers to what?** Amplitude is fixed. Intensity is coded by frequency.
4. **Why can’t you fire during absolute refractory?** Na+ channels are inactivated, not just closed.
5. **What causes hyperpolarization?** K+ channels stay open after Na+ has stopped.

6. **Saltatory conduction needs what?** Myelin + Nodes of Ranvier with Na+ channels.
7. **Fastest fiber type?** A-alpha (large, myelinated, motor + proprioception).
8. **Slowest fiber type?** C (small, unmyelinated, slow pain, postganglionic autonomic).
9. **Where does the AP start?** Axon hillock (lowest threshold, most Na+ channels).
10. **Na+/K+ pump ratio?** 3 Na+ out, 2 K+ in, per 1 ATP. Electrogenic.

8. Quick-recall summary

Question	Answer
RMP value	-70 mV
Threshold	-55 mV
Peak voltage	+30 mV
Hyperpolarized to	-80 mV
Set by	K+ leak channels
Depolarization caused by	Na+ in (voltage-gated)
Repolarization caused by	K+ out (voltage-gated)
Hyperpolarization caused by	K+ still open
Returns to RMP via	Na+/K+ pump (3 Na+ out, 2 K+ in)
AP triggered at	Axon hillock
Transmitter release triggered by	Ca2+ entry at terminal
Absolute refractory caused by	Na+ channel inactivation
Relative refractory caused by	K+ still open (hyperpolarized)
Fastest conduction	Saltatory in myelinated A-alpha (up to 120 m/s)

9. 10 MET-style MCQs

Q1. The resting membrane potential of a typical neuron is closest to: A. -30 mV B. -55 mV C. -70 mV D. -90 mV
Answer
C. -70 mV. A is too depolarized, B is threshold, D is K+ equilibrium potential.
Q2. Resting membrane potential is primarily determined by: A. Na+/K+ ATPase B. K+ leak channels C. Voltage-gated Na+ channels D. Cl- channels
Answer
B. K+ leak channels. The pump maintains gradients but is not the direct voltage-setter. Common trap.
Q3. During depolarization phase of an AP, the dominant ion movement is: A. K+ out B. Na+ in C. Ca2+ in D. Cl- in
Answer
B. Na+ in. Voltage-gated Na+ channels open at threshold.
Q4. Repolarization of a neuron is caused mainly by: A. Closing of K+ channels B. Opening of voltage-gated K+ channels C. Na+/K+ pump activity D. Ca2+ entry
Answer
B. K+ rushes out down its gradient.

Q5. The Na+/K+ ATPase moves: A. 2 Na+ out, 3 K+ in B. 3 Na+ out, 2 K+ in C. 3 Na+ in, 2 K+ out D. 1 Na+ out, 1 K+ in
Answer
B. 3 Na+ out, 2 K+ in. Electrogenic (net +1 export).
Q6. During the absolute refractory period: A. A stronger stimulus can elicit AP B. Only Cl- channels are open C. Na+ channels are inactivated D. K+ channels are closed
Answer
C. Inactivated, not just closed. No AP possible.
Q7. Saltatory conduction requires: A. Unmyelinated axons B. Continuous Na+ channels along the axon C. Nodes of Ranvier D. C fibers
Answer
C. Nodes of Ranvier with concentrated Na+ channels.
Q8. The all-or-none principle states that: A. AP amplitude varies with stimulus strength B. AP amplitude is fixed once threshold is reached C. APs do not propagate D. APs are graded
Answer
B. Stimulus intensity is encoded as frequency, not amplitude.
Q9. Neurotransmitter release at the axon terminal is triggered by: A. Na+ entry B. K+ exit C. Ca2+ entry D. Cl- entry
Answer
C. Ca2+ entry through voltage-gated Ca2+ channels at the terminal.
Q10. An IPSP typically involves: A. Na+ influx B. Cl- influx or K+ efflux C. Ca2+ influx D. Closing of K+ channels
Answer
B. Cl- in or K+ out hyperpolarizes the cell, moves it away from threshold.

10. Memorization aids

- **“Na+ in, K+ out”** — the AP in 4 words.
- **“3 out, 2 in, 1 ATP”** — the pump.
- **“Ca for Communication”** — Ca2+ at the terminal triggers transmitter release.
- **“Inactivated, not closed”** — why absolute refractory exists.
- **“Salt jumps Nodes”** — saltatory = jumps between Nodes of Ranvier.
- **“K sets the rest”** — RMP is a K+ phenomenon.

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Sleep — Full Primer

Sleep, Full Primer

Built 2026-04-22 after Sihan reported “I do not know anything about sleep.” This covers everything MET might ask, organized from macro (architecture) to micro (neurochemistry) to clinical (disorders). Memorize the tables.

1. Macro View: Sleep is Two Modes

Sleep splits into **NREM (Non-REM)** and **REM (Rapid Eye Movement)**. NREM has 3 stages. Total = 4 stages of sleep.

	NREM	REM
Brain activity	Slows progressively	Fast, almost like awake
Muscle tone	Normal, relaxed	**Atonia (paralyzed)**
Eye movements	Slow or absent	Rapid bursts
Dreaming	Rare, thought-like	**Vivid, narrative**
Heart rate / BP	Low, stable	Variable, elevated
Breathing	Slow, regular	Irregular
Thermoregulation	Works	**Fails** (we do not shiver/sweat normally)
Memory role	Declarative consolidation (esp. NREM 3)	Procedural + emotional consolidation

2. The Four Stages

NREM Stage 1 — Transition (about 5% of night)

- EEG: **theta waves (4–8 Hz)**.
- Very light sleep, easily woken. People often say they were not asleep.
- Hypnagogic images, muscle twitches (hypnic jerks).
- Duration: 5–10 minutes at start of night.

NREM Stage 2 — Light Sleep (about 45–55% of night)

- EEG: theta + **sleep spindles** (12–14 Hz bursts) + **K-complexes** (large single waves).
- Core sign of being “truly asleep.” Body temperature drops, heart rate slows.
- Most of the night spent here. Increases as night progresses.
- Spindles come from the thalamus, gate sensory input so you stay asleep.

NREM Stage 3 — Slow-Wave Sleep / Deep Sleep / Delta (about 15–20%)

- EEG: **delta waves (0.5–4 Hz), high amplitude**.
- Deepest sleep. Hard to rouse. If woken, groggy (sleep inertia).
- **Growth hormone** released here. Immune repair, glycogen restoration.

- **Parasomnias** (sleepwalking, sleep terrors, bedwetting) come OUT of NREM 3, not REM. Common in children because children have more delta sleep.
- Dominates the FIRST HALF of the night.

REM — Paradoxical Sleep (about 20–25%)

- EEG: **mixed frequency, low amplitude**, looks almost like awake. "Paradoxical" because brain is active but body is paralyzed.
- **Muscle atonia** (pons actively suppresses motor output via medial medulla → spinal cord). Only eye muscles and diaphragm keep moving.
- **Vivid narrative dreams.** Emotional, story-like.
- Eye bursts (the "RE" in REM).
- Autonomic instability: irregular HR, BP, breathing. Penile/clitoral tumescence.
- REM episodes GET LONGER across the night. First REM is short (10 min), last REM can be 45+ min.
- Dominates the SECOND HALF of the night.

3. Sleep Cycle Architecture

One full cycle = **~90 minutes**. Sequence through the night, typical young adult:

```
Night start:
Wake → N1 → N2 → N3 → N2 → REM → (first cycle, ~90 min, mostly N3, short REM)
     → N2 → N3 → N2 → REM → (second cycle, less N3, longer REM)
     → N2 → REM       → (third cycle, minimal N3, long REM)
     → N2 → REM       → (fourth / fifth cycles: REM-dominated, minimal N3)
```

- 4–6 cycles per 8 hours.
- First half of night = deep sleep heavy (N3 for body repair).
- Second half = REM heavy (for memory consolidation and dreams).
- If sleep is cut short, you lose REM disproportionately.

4. Neural Control

Sleep onset

- Promoted by **adenosine** accumulating through the day (caffeine blocks adenosine receptors, which is why coffee keeps you awake).
- **VLPO (Ventrolateral preoptic nucleus)** of the hypothalamus = "sleep switch." GABAergic, inhibits wake-promoting regions.
- **Melatonin** from pineal gland, released in darkness.

Wake

- **Ascending Reticular Activating System (ARAS)** of the brainstem.

- Neurotransmitters: histamine (tuberomammillary nucleus), orexin/hypocretin (lateral hypothalamus), norepinephrine (locus coeruleus), serotonin (raphe), acetylcholine (PPT/LDT).
- Loss of orexin neurons = **narcolepsy**.

REM vs NREM switching

- Driven by a flip-flop between:
  - **REM-ON cells:** cholinergic neurons in PPT/LDT (acetylcholine surge during REM).
  - **REM-OFF cells:** aminergic neurons (NE in locus coeruleus, 5-HT in raphe) — silent during REM.
- Atonia mechanism: during REM, pontine REM-on neurons activate medial medulla (magnocellular nucleus) → hyperpolarizes spinal motor neurons → full-body paralysis except diaphragm and eyes.

Circadian

- **SCN (Suprachiasmatic nucleus)** of the hypothalamus = master clock, ~24.2 hr intrinsic period.
- Entrained by **light** via the retinohypothalamic tract (non-visual, from intrinsically photosensitive retinal ganglion cells).
- SCN → pineal gland → **melatonin** release in darkness.
- Core body temperature dips at night (~2 AM), rises in morning.

5. Sleep Across the Lifespan

Age	Total sleep	% REM	Notes
Newborn	16–18 hrs	~50%	Enters sleep via REM (adults enter via N1)
Infant (1 yr)	14 hrs	~30%	—
Child (5 yr)	11–12 hrs	~25%	Most parasomnias occur now
Adolescent	9 hrs	~20%	Delayed sleep phase is developmentally normal
Adult	7–8 hrs	~20–25%	Standard baseline
Elderly (65+)	6–7 hrs	~15–20%	Less N3, more awakenings, phase advance

6. Sleep Disorders (DSM-5, MCQ High Yield)

Dyssomnias (difficulty with amount/timing/quality)

Disorder	Core feature	Treatment
Insomnia	Difficulty falling/staying asleep, 3 nights/wk × 3 months	CBT-I first line, then hypnotics short-term
Narcolepsy	Excessive daytime sleepiness, **cataplexy** (sudden muscle tone loss triggered by emotion), **sleep paralysis**, **hypnagogic/ hypnopompic hallucinations**. Cause: **orexin/hypocretin deficiency**. REM intrudes into wake.	Modafinil, stimulants; sodium oxybate for cataplexy
Obstructive Sleep Apnea (OSA)	Airway collapses during sleep, repeated apneas/hypopneas, snoring, daytime sleepiness	CPAP
Central Sleep Apnea	Brainstem fails to signal breathing	CPAP/BiPAP
Restless Legs Syndrome	Urge to move legs, worse at rest/night, relieved by movement. Dopamine deficit, low iron	Dopamine agonists (pramipexole), iron
Circadian Rhythm Disorders	Delayed sleep phase, advanced sleep phase, shift work, jet lag	Light therapy, melatonin
Kleine-Levin Syndrome	Recurrent episodes of hypersomnia + hyperphagia + hypersexuality. Young males	Rare

Parasomnias (abnormal events during sleep)

Disorder	Stage	Core feature
Sleepwalking (somnambulism)	**NREM 3**	Complex motor activity, eyes open, no memory. Children.
Sleep terrors (night terrors)	**NREM 3**	Sudden arousal with scream, autonomic storm, no memory. Children.
Nightmares	**REM**	Vivid scary dream, full recall on waking. Any age.
REM Sleep Behavior Disorder (RBD)	**REM**	**Loss of atonia**. Patient acts out dreams (punching, yelling). Associated with α-synucleinopathies (Parkinson, Lewy body dementia). Older males.
Sleep paralysis	**REM onset/ offset**	Brief atonia while awake, terrifying. Can occur in narcolepsy or isolated.

Disorder	Stage	Core feature
Bruxism	Any stage	Teeth grinding
Enuresis (bedwetting)	**NREM 3**	Children. Usually resolves with age.

7. Sleep and Memory

- **Declarative / explicit memory** (facts, events) consolidated during **NREM** (especially slow-wave sleep). Hippocampus replays during delta.
- **Procedural / implicit memory** (motor skills) consolidated during **REM** and **Stage 2 (spindles)**.
- **Emotional memory** consolidated during **REM** (amygdala active, prefrontal cortex quiet → emotional tagging without rational evaluation).
- Sleep deprivation hurts consolidation disproportionately.

8. Sleep and Deprivation

- **Total sleep deprivation (24 hr):** impaired vigilance, slowed reaction time, mood lability, microsleeps.
- **Chronic partial sleep restriction** is as bad as acute total deprivation over time, but people feel less impaired (adaptation perception).
- **REM rebound** after REM deprivation: next night's sleep has more REM to "catch up." Delta rebound similarly.
- **Sleep deprivation paradoxically treats depression** short-term (via increased BDNF, reduced REM intrusion), though benefit resolves after sleep.

9. MCQ Trap Cluster (memorize cold)

1. **Muscle atonia → REM.** Always. Not delta, not stage 2.
2. **Sleep spindles + K-complexes → NREM 2.** Signature feature.
3. **Delta waves + slow-wave sleep + "deep sleep" → NREM 3.**
4. **Paradoxical sleep = REM.** Brain awake, body asleep.
5. **Parasomnias (sleepwalking, night terrors, enuresis) → NREM 3**, not REM.
6. **RBD (REM sleep behavior disorder) = loss of atonia in REM.** Elderly men, α-synucleinopathy warning sign.
7. **Narcolepsy = orexin/hypocretin deficiency.** Cataplexy is pathognomonic.
8. **SCN = master circadian clock**, entrained by **light via retinohypothalamic tract**.
9. **Melatonin from pineal gland**, released in **darkness**.
10. **Adenosine drives sleep pressure; caffeine blocks adenosine receptors.**
11. **REM-ON neurons = cholinergic (PPT/LDT).** REM-OFF = aminergic (NE, 5-HT).
12. **Newborns enter sleep via REM** (active sleep), adults enter via NREM 1.
13. **N3 dominates first half of night; REM dominates second half.**
14. **Declarative memory → NREM (delta). Procedural → REM + spindles.**
15. **Sleep cycle = ~90 minutes.** 4–6 cycles per night.
16. **Growth hormone released in NREM 3.**
17. **Pons + medial medulla → atonia during REM** (hyperpolarize spinal motor neurons).

18. **Temperature regulation fails during REM** (poikilothermic).
19. **Kleine-Levin** = hypersomnia + hyperphagia + hypersexuality (the three H's).
20. **Insomnia DSM-5 threshold:** 3 nights/week for 3 months.

10. Flashcard-Ready Facts

If a question says … → answer is … - "Paralysis during sleep" → REM - "Sleep spindles" → NREM 2 - "K-complexes" → NREM 2 - "Delta waves" → NREM 3 - "Slow-wave sleep" → NREM 3 - "Paradoxical sleep" → REM - "Growth hormone secreted in" → NREM 3 - "Acetylcholine peaks in" → REM - "Orexin deficiency" → Narcolepsy - "Cataplexy" → Narcolepsy - "Sudden screaming with autonomic arousal in a 7-year-old, no memory" → Sleep terrors (NREM 3) - "Acting out dreams in a 68-year-old male" → RBD (REM) - "Urge to move legs worse at rest" → RLS - "Cortex looks asleep, body is moving" → sleepwalking (NREM 3) - "Cortex looks awake, body is paralyzed" → REM - "Master circadian clock" → SCN - "Melatonin secreted by" → Pineal gland - "Sleep onset driven by" → Adenosine + VLPO - "Caffeine mechanism" → Adenosine receptor blocker - "Sleep cycle length" → ~90 minutes

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Brain Anatomy

Brain — Anatomy, Parts, Functions

Full MET-aligned reference. Every clinical association listed is fair game for MCQs.

1. Overall Organization

Forebrain (Prosencephalon): - Telencephalon → cerebral hemispheres + basal ganglia + limbic cortex - Diencephalon → thalamus + hypothalamus + epithalamus (pineal) + subthalamus

Midbrain (Mesencephalon): - Tectum (dorsal): superior colliculi (visual reflex) + inferior colliculi (auditory relay) - Tegmentum (ventral): VTA (dopamine), substantia nigra, red nucleus, periaqueductal gray

Hindbrain (Rhombencephalon): - Metencephalon → pons + cerebellum - Myelencephalon → medulla oblongata

Spinal cord continues from medulla.

2. Cerebral Cortex — 4 Lobes

Frontal Lobe

- **Primary motor cortex (M1)** — precentral gyrus, BA4. Somatotopic map = motor homunculus. Contralateral control.
- **Premotor + supplementary motor area** — planning movement sequences.
- **Broca's area** — L inferior frontal gyrus, BA44/45. Speech production.
- **Prefrontal cortex (PFC):**
  - Dorsolateral PFC — working memory, executive function, planning
  - Ventromedial PFC — personality, emotion regulation, decision-making (Phineas Gage)
  - Orbitofrontal cortex — impulse control, reward processing
- **Frontal eye fields** — voluntary eye movements

Lesions: - Motor cortex → contralateral paralysis/paresis - Broca's → Broca's aphasia (non-fluent, effortful, agrammatic, good comprehension) - Dorsolateral PFC → dysexecutive syndrome (WCST perseverative errors) - Ventromedial PFC → disinhibition, poor judgment, flat affect (Phineas Gage) - Frontal lobe (generally) → primitive reflexes return (grasp, snout, palmomental)

Parietal Lobe

- **Primary somatosensory cortex (S1)** — postcentral gyrus, BA3/1/2. Sensory homunculus.
- **Superior parietal lobule** — spatial attention, visuomotor
- **Inferior parietal lobule (R):** body schema, attention to contralateral space
- **Inferior parietal lobule (L):** includes angular + supramarginal gyri. Language-adjacent.

Lesions: - S1 → contralateral sensory loss - R parietal → **hemispatial neglect** (ignores L side), dressing apraxia, anosognosia - L angular gyrus → **Gerstmann syndrome** (agraphia, acalculia, finger agnosia, L-R confusion) - Astereognosis (can't recognize object by touch) - Balint's syndrome (bilateral parietal) — simultanagnosia, optic ataxia, oculomotor apraxia

Temporal Lobe

- **Primary auditory cortex (A1)** — Heschl's gyrus, BA41/42
- **Wernicke's area** — L posterior superior temporal gyrus, BA22. Language comprehension.
- **Hippocampus** (medial) — declarative memory encoding
- **Amygdala** (anteromedial) — fear, emotional learning
- **Fusiform gyrus** — face recognition, word form

Lesions: - A1 → cortical deafness (bilateral) - Wernicke's → **Wernicke's aphasia** (fluent, nonsensical, poor comprehension, word salad) - Bilateral hippocampus → **anterograde amnesia** (H.M.), preserved procedural memory - Fusiform gyrus → **prosopagnosia** (face blindness) - Bilateral amygdala → **Klüver-Bucy syndrome**

(hyperorality, hypersexuality, placidity, visual agnosia) - L temporal → anomia - Temporal lobe epilepsy → déjà vu, jamais vu, olfactory/auditory hallucinations, religious experiences

Occipital Lobe

- **Primary visual cortex (V1)** — BA17, calcarine fissure, striate cortex (from stripe of Gennari)
- **V2, V3, V4** — color (V4), motion (V5/MT)
- **Ventral stream** (V1 → temporal): "what" — object/face recognition
- **Dorsal stream** (V1 → parietal): "where/how" — spatial, action

Lesions: - V1 → cortical blindness + blindsight (can still react without conscious vision) - V4 → achromatopsia (color blindness central) - V5/MT → akinetopsia (can't see motion) - Ventral stream → visual agnosia (Lissauer: apperceptive vs associative)

3. Hemisphere Specialization

Left hemisphere (in most right-handers): - Language (Broca + Wernicke) - Logic, sequential processing, math - Analytical

Right hemisphere: - Spatial, visuospatial - Music, prosody (emotional tone of speech) - Face recognition (fusiform) - Holistic processing - Emotion (especially negative)

Corpus callosum — 200M fibers, largest commissure. Splits = **split-brain syndrome** (Sperry/Gazzaniga): each hemisphere acts independently, alien hand, can't name objects presented to L visual field (in R hemisphere → can't reach language L hemisphere).

Anterior commissure — smaller, connects temporal lobes.

4. Subcortical Structures

Thalamus

- **Sensory relay** for ALL senses except olfaction.
- Key nuclei:
  - **LGN** (lateral geniculate) — vision
  - **MGN** (medial geniculate) — audition
  - **VPL/VPM** — somatosensation (body / face)
  - **Ventral anterior, ventral lateral** — motor
  - **Mediodorsal** — to PFC, memory, emotion
  - **Pulvinar** — attention
- Gateway to cortex. Damage → sensory loss + cognitive deficits.

Hypothalamus

- Master **homeostatic** control. Small (4 g) but critical.

- Key nuclei:
  - **Lateral (LH)** — hunger ("on" for eating; lesion → aphagia)
  - **Ventromedial (VMH)** — satiety ("off" for eating; lesion → hyperphagia + obesity)
  - **Paraventricular (PVN)** — CRH, oxytocin, ADH
  - **Supraoptic** — ADH, oxytocin
  - **Suprachiasmatic (SCN)** — master circadian clock
  - **Preoptic area** — thermoregulation; sexual behavior (sexually dimorphic nucleus)
  - **Arcuate nucleus** — appetite integration (leptin, ghrelin signals), pituitary control
- Controls pituitary → endocrine system.
- Controls ANS (sympathetic: posterior; parasympathetic: anterior).

Basal Ganglia

- Motor control + habit learning + reward.
- Components:
  - **Striatum** = caudate + putamen (input)
  - **Globus pallidus** (external + internal segments) (output)
  - **Substantia nigra** (pars compacta — dopamine to striatum; pars reticulata — output)
  - **Subthalamic nucleus**
  - **Nucleus accumbens** (part of ventral striatum) — reward, addiction
- **Direct pathway** (D1 receptors) — promotes movement
- **Indirect pathway** (D2 receptors) — suppresses movement

Disorders: - **Parkinson's disease** — substantia nigra dopamine neurons degenerate → rigidity, bradykinesia, resting tremor, postural instability. Treatment: L-DOPA. - **Huntington's disease** — caudate/putamen degeneration → chorea, cognitive decline, psychiatric symptoms. Autosomal dominant, CAG repeats chr 4. - **Tourette's** — basal ganglia dysfunction → motor + vocal tics - **OCD** — implicated in basal ganglia-thalamic-cortical loops - **Hemiballismus** — subthalamic nucleus lesion

Limbic System

Emotion + memory. Papez circuit (1937): **Hippocampus → fornix → mammillary bodies → anterior thalamus → cingulate gyrus → parahippocampal gyrus → hippocampus**

Components: - **Hippocampus** — declarative memory consolidation. CA1 = site of LTP. Sea-horse shape. - **Amygdala** — fear, emotional memory, social/emotional processing - **Cingulate gyrus** (ant + post) — emotion, attention, pain, conflict monitoring - **Parahippocampal gyrus** — context, spatial memory - **Mammillary bodies** — memory (damaged in Korsakoff's) - **Septal nuclei** — reward - **Fornix** — hippocampus output

Cerebellum

- 10% of brain volume, **50% of neurons** (Purkinje, granule).
- Lobes: anterior (posture), posterior (coordination), flocculonodular (balance, eye movements).

- **Cerebrocerebellum** (lateral) — motor planning
- **Spinocerebellum** (medial vermis + paravermis) — execution
- **Vestibulocerebellum** (flocculonodular) — balance, VOR

Functions: - Coordination, timing, precision of movement - Procedural learning (classical conditioning of motor responses, e.g., eyeblink) - Increasingly recognized role in cognition (cerebellar cognitive affective syndrome)

Lesions: - Ataxia (gait) - Dysmetria (over/undershooting) - **Intention tremor** (worsens as target approached — contrast with Parkinson's resting tremor) - Dysdiadochokinesia (can't do rapid alternating movements) - Nystagmus - Scanning speech

Brainstem

Top to bottom:

Midbrain: - Tectum — superior colliculus (visual reflex, saccades), inferior colliculus (auditory relay) - Tegmentum — red nucleus (rubrospinal tract), substantia nigra (DA), VTA (DA), periaqueductal gray (pain modulation, defensive behavior) - Cerebral peduncles — descending motor tracts

Pons: - Pontine nuclei — cerebellum relay - Locus coeruleus — NE source (arousal, attention) - Raphe nuclei — 5-HT source - Reticular formation (continues) - Cranial nerve nuclei V-VIII

Medulla oblongata: - Cardiovascular + respiratory centers (vital) - Pyramids (decussation of motor tracts — why L brain controls R body) - Olivary nuclei - Cranial nerve nuclei IX-XII - Area postrema — vomiting (outside BBB)

Reticular Activating System (RAS): diffuse network through brainstem, regulates arousal, consciousness, sleep-wake cycle. Damage → coma.

5. Ventricles & CSF

Lateral (2) → Interventricular foramen of Monro → 3rd → Aqueduct of Sylvius → 4th → central canal / subarachnoid space

- CSF produced by **choroid plexus** (~500 mL/day, total ~150 mL).
- Absorbed by **arachnoid granulations** into dural venous sinuses.
- **Hydrocephalus** — CSF accumulation (communicating vs non-communicating).
- **Normal pressure hydrocephalus** — triad of dementia, gait disturbance, urinary incontinence.

6. Brodmann Areas (memorize these)

BA	Region	Function
1, 2, 3	Postcentral gyrus	Primary somatosensory (S1)
4	Precentral gyrus	Primary motor (M1)

BA	Region	Function
6	Frontal	Premotor + SMA
8	Frontal	Frontal eye fields
17	Occipital	Primary visual (V1), striate
18, 19	Occipital	Visual association
22	Superior temporal	Wernicke's (L), auditory association
39	Angular gyrus	Reading, math (Gerstmann)
40	Supramarginal	Language, phonology
41, 42	Heschl's gyrus	Primary auditory (A1)
44, 45	Inferior frontal	Broca's (L), pars opercularis + triangularis
46	DLPFC	Working memory, executive

7. Arterial Supply (fast coverage)

Circle of Willis: - Anterior cerebral artery (ACA) — medial frontal + parietal (leg > arm) - Middle cerebral artery (MCA) — lateral cortex (face + arm > leg, language areas) - Posterior cerebral artery (PCA) — occipital + medial temporal - Basilar → pons, midbrain, cerebellum

Classic stroke syndromes: - MCA L → Broca + Wernicke aphasia, R hemiplegia (face + arm) - MCA R → L hemispatial neglect - ACA → contralateral leg > arm weakness - PCA → contralateral hemianopia

8. High-Yield Clinical Pearls

- **Wernicke-Korsakoff syndrome** — thiamine deficiency (alcoholism). Wernicke's = acute (ataxia, confusion, ophthalmoplegia). Korsakoff's = chronic amnesia + confabulation. Mammillary bodies + thalamus damage.
- **Capgras delusion** — belief familiar person is replaced by impostor. Temporoparietal damage.
- **Prosopagnosia** — fusiform face area (FFA) damage.
- **Alien hand syndrome** — supplementary motor area or callosal damage.
- **Locked-in syndrome** — ventral pons lesion. Intact consciousness + eye movement, total body paralysis.
- **Persistent vegetative state** — cortex dead, brainstem intact.
- **Alzheimer's disease** — plaques (beta-amyloid) + tangles (tau). Starts in entorhinal cortex + hippocampus. ACh ↓ (basal forebrain, nucleus basalis of Meynert).

Memorization Hooks

- **M → P → T → O**: Motor (frontal), Parietal, Temporal, Occipital — anterior to posterior
- **“Precentral = Motor, Postcentral = Sensory”** — remember the “central sulcus” separates them
- Broca = **frontal = speaks broken** (non-fluent)
- Wernicke = **temporal = wordy but wrong** (fluent)
- **LGN = Light, MGN = Music** (lateral = vision, medial = audition)
- Basal ganglia **“Parkinson’s too slow, Huntington’s too fast”**
- Hippocampus + Amygdala = **“memory + mood”** twins in medial temporal lobe

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Neurons and Glia

Neurons and Glial Cells — Complete MET Reference

Every MCQ-testable cell type in the nervous system. Emphasis on the traps examiners love.

1. Neuron Classification

By Structure (morphology)

Type	Processes	Where Found	Example
Multipolar	1 axon + MANY dendrites	Most of CNS, motor neurons	Pyramidal cells (cortex), spinal motor neurons, Purkinje cells
Bipolar	1 axon + 1 dendrite	Special sensory	Retina, olfactory epithelium, cochlear + vestibular ganglia
Unipolar / Pseudounipolar	Single process that branches	Sensory ganglia	Dorsal root ganglia (touch/pain/proprio to spinal cord)
Anaxonic	No distinguishable axon	Select CNS	Amacrine cells (retina), some interneurons

Most common type in humans: multipolar.

By Function

Type	Direction	Role
Sensory (afferent)	Periphery → CNS	Carry sensation in
Motor (efferent)	CNS → muscle/gland	Carry commands out
Interneurons	CNS internal	Connect sensory to motor; most of the brain

Most common type in CNS: interneurons (~99% of CNS neurons).

Special Named Neurons (MET likes these)

- **Pyramidal cells** — cortex (layers III and V), triangular soma, long apical dendrite. Main output neurons.
- **Purkinje cells** — cerebellar cortex, massive dendritic tree (one of largest in CNS). Output of cerebellar cortex.
- **Granule cells** — cerebellum + dentate gyrus of hippocampus. Smallest + most numerous in brain.
- **Basket cells** — inhibitory interneurons, cerebellum and cortex.
- **Betz cells** — giant pyramidal cells in M1 (primary motor cortex), fastest conducting.
- **Mirror neurons** — fire both when performing action AND observing same action (first found in monkey F5 area, proposed role in imitation, empathy, language evolution — Rizzolatti).

2. Glial Cells — The 6 Types You Must Know

Glia (Greek "glue") outnumber neurons approximately 1:1 in human brain (the old 10:1 figure is outdated). They don't fire APs but are essential.

CNS Glia (4 types — all from neuroectoderm except microglia)

2.1 Astrocytes (star-shaped; most abundant CNS glia) - Form **blood-brain barrier (BBB)** via end-feet wrapping capillaries + tight junctions between endothelial cells - **K^+ buffering** — soak up extracellular K^+ released during neuronal firing - **Glutamate recycling** — remove glutamate from synaptic cleft, convert to glutamine, return to neurons - Metabolic support (glucose → lactate shuttle to neurons) - **Reactive astrogliosis** — form glial scar after CNS injury (barrier, but also inhibits axonal regrowth — one reason CNS doesn't regenerate like PNS) - Subtypes: **protoplasmic** (gray matter, bushy) vs **fibrous** (white matter, long processes)

2.2 Oligodendrocytes (few branches) - Produce **myelin in CNS** - **ONE oligodendrocyte myelinates MULTIPLE axons** (up to ~50 internodal segments) - Damaged in **Multiple Sclerosis (MS)** — autoimmune demyelination of CNS → scattered plaques, varied neurological deficits - Also damaged in progressive multifocal leukoencephalopathy (PML, JC virus in immunocompromised)

2.3 Microglia (small; "resident macrophages of CNS") - CNS **immune cells** — phagocytosis, antigen presentation, cytokine release - **EXCEPTION: derived from mesoderm/myeloid** (yolk sac origin), NOT neuroectoderm like other glia — a favorite MCQ trap - Activated in infection, neurodegeneration (Alzheimer's, Parkinson's), stroke, trauma - Prune synapses during development + adulthood

2.4 Ependymal cells (cuboidal, ciliated) - Line the **ventricular system** and central canal of spinal cord - **Choroid plexus** = modified ependymal cells that produce **CSF** (~500 mL/day, total volume ~150 mL) - Cilia circulate CSF

PNS Glia (2 types)

2.5 Schwann cells (neurilemmocytes) - Produce **myelin in PNS** - **ONE Schwann cell myelinates ONE internodal segment of ONE axon** (opposite ratio from oligos) - Form **neurilemma** (cytoplasmic sheath outside myelin) — critical for **axonal regeneration** in PNS (which is why PNS can regrow, CNS cannot) - Damaged in **Guillain-Barré syndrome** (acute autoimmune demyelination, ascending paralysis) - Also damaged in Charcot-Marie-Tooth disease (hereditary)

2.6 Satellite cells - Surround neuron cell bodies in **PNS ganglia** (dorsal root ganglia, autonomic ganglia) - Provide support, regulate microenvironment, buffer K^+ and NTs - Analogous to astrocytes but in PNS

3. Side-by-Side Comparisons (high-yield MCQ territory)

Oligodendrocyte vs Schwann cell

Feature	Oligodendrocyte	Schwann cell
Location	CNS	PNS
Axons myelinated per cell	**MANY (up to ~50)**	**ONE segment of ONE axon**
Disease	Multiple Sclerosis, PML	Guillain-Barré, Charcot-Marie-Tooth
Regeneration support	No (CNS doesn't regenerate well)	Yes (neurilemma guides regrowth)
Embryonic origin	Neuroectoderm	Neural crest

Astrocyte vs Microglia

Feature	Astrocyte	Microglia
Abundance	Most abundant CNS glia	~10-15% of CNS glia
Function	Support, BBB, K^+, glutamate recycling	Immune, phagocytosis
Origin	Neuroectoderm	Mesoderm (yolk sac) ← TRAP

Feature	Astrocyte	Microglia
Activation	Reactive astrogliosis (scar)	Activated microgliosis (cytokines)

Protoplasmic vs Fibrous Astrocytes

Feature	Protoplasmic	Fibrous
Location	Gray matter	White matter
Shape	Bushy, short processes	Long, thin processes
Function	Local synaptic support	Axon tract support

4. Myelination Details

- **Myelin** = concentric wrapping of glial cell plasma membrane around axon, ~70-80% lipid (hence white appearance of white matter)
- **Nodes of Ranvier** = unmyelinated gaps between myelin segments; site of voltage-gated Na^+ channels; AP "jumps" node to node = **saltatory conduction** (fast)
- **Unmyelinated axons** → continuous conduction (slow)
- Conduction speed depends on: axon **diameter** (larger = faster) + **myelination** (more = faster)
- Fastest conducting: large myelinated fibers (e.g., Aα motor neurons, 120 m/s)
- Slowest: small unmyelinated (e.g., C fibers for dull pain, 0.5-2 m/s)

Fiber Classifications (Erlanger-Gasser)

Class	Diameter	Myelin	Speed	Function
Aα	Largest	Heavy	80-120 m/s	Motor, proprioception
Aβ	Large	Heavy	35-75 m/s	Touch, pressure
Aγ	Medium	Moderate	15-30 m/s	Muscle spindle motor
Aδ	Small	Light	5-30 m/s	Fast pain, temperature
B	Small	Light	3-15 m/s	Preganglionic autonomic
C	Smallest	**None**	0.5-2 m/s	Slow pain, postganglionic autonomic

5. High-Yield MCQ Traps (memorize these exact facts)

1. **Most abundant glia in PNS → Schwann cells** (appeared on Sihan's Phase 1). PNS has only 2 glial types (Schwann + satellite). Schwann cells wrap every PNS axon — myelinating OR non-myelinating (Remak bundles for unmyelinated axons). Satellite cells are only in ganglia. So Schwann >> Satellite.
2. **Most abundant glia in CNS → Astrocytes** (not oligodendrocytes)
3. **Glia that forms myelin in CNS → Oligodendrocyte**
4. **Glia that forms myelin in PNS → Schwann cell**
5. **One Schwann** = **one segment of one axon**. **One oligodendrocyte** = **many axons**.
6. **Microglia** come from **mesoderm/myeloid** lineage, not neuroectoderm — distinct from all other CNS glia.
7. **CSF producer → Choroid plexus** (modified ependymal cells). Not astrocytes, not arachnoid cells.
8. **Blood-brain barrier** → formed by astrocyte end-feet + tight junctions of **capillary endothelial cells** (not neurons, not microglia).
9. **MS** damages **oligodendrocytes** (CNS demyelination).
10. **Guillain-Barré** damages **Schwann cells** (PNS demyelination).
11. **PNS regenerates, CNS doesn't** — because Schwann cells form a neurilemma that guides regrowth, while reactive astrocytes form an inhibitory glial scar.
12. **Purkinje cells** are the only output of the cerebellar cortex. They are **inhibitory** (GABAergic).
13. **Betz cells** in M1 are giant pyramidal neurons, fastest motor conduction.
14. **Mirror neurons** — Rizzolatti, monkey F5. Human equivalent: inferior frontal gyrus + inferior parietal lobule.
15. **Granule cells** are the most numerous neurons in the brain (mostly in cerebellum and hippocampus DG).
16. **Glia : neuron ratio in human brain ≈ 1:1** (old 10:1 figure is outdated).

6. Quick Mnemonics

- **"Oligos handle many, Schwanns one"** — ratio difference
- **"Microglia are Mesodermal"** — both start with M
- **"Ependymal cilia push CSF"**
- **"Astrocyte ends the vessel"** (end-feet → BBB)
- **"MS = Many Schwanns safe"** (MS is CNS/oligo only; Schwann cells are untouched)
- **"GBS = Guillain Beats Schwann"** (mnemonic for PNS demyelination)

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Biopsychology Master Cheatsheet

Biopsychology — Master Cheat Sheet (MET 2026)

Single-pass reference for one-hour review sessions. Every fact here is MET-MCQ territory.

1. Neuron & Neural Communication

Neuron parts: dendrites (receive) → soma (integrate) → axon hillock (trigger) → axon (conduct) → terminal buttons (release NT) **Myelin:** Schwann cells (PNS), oligodendrocytes (CNS). Gaps = Nodes of Ranvier. Conduction = **saltatory** (jumps node to node). **Resting potential:** −70 mV. Maintained by Na^+ /K^+ pump (3 Na^+ out, 2 K^+ in) + leaky K^+ channels. **Action potential (AP):** threshold ≈ −55 mV → Na^+ in (depolarize to +30) → K^+ out (repolarize) → hyperpolarize → resting. **All-or-none** + **absolute refractory period** (no AP possible) + **relative refractory** (needs stronger stimulus). **Speed:** larger axon + more myelin = faster. Unmyelinated = continuous conduction (slow).

Synapse types: - **Chemical** (most): AP → Ca^{2+} in → vesicles fuse → NT released → binds postsynaptic receptors. Slow (~0.5-2 ms). Modifiable (learning). - **Electrical** (gap junctions): direct ion flow. Ultra-fast. Bidirectional. Not modifiable.

EPSP vs IPSP: Excitatory postsynaptic potential = depolarizing (Na^+ in). Inhibitory = hyperpolarizing (Cl^- in or K^+ out). Summation (spatial, temporal) at axon hillock decides if AP fires.

2. Neurotransmitters (know all 7 families)

NT	Role / Key Facts	Disorders
Acetylcholine (ACh)	Muscle, memory, attention. NMJ. Parasympathetic	↓ in Alzheimer's (basal forebrain/nucleus basalis of Meynert)
Dopamine (DA)	Reward, motivation, motor control. Pathways: mesolimbic, mesocortical, nigrostriatal, tuberoinfundibular	↑ in schizophrenia (mesolimbic), ↓ in Parkinson's (substantia nigra → striatum)
Serotonin (5-HT)	Mood, sleep, appetite. Raphe nuclei (brainstem)	↓ in depression (SSRIs), OCD, anxiety
Norepinephrine (NE/ NA)	Arousal, fight-or-flight, attention. Locus coeruleus	↓ in depression, ↑ in anxiety/PTSD
GABA		↓ in anxiety, epilepsy

NT	Role / Key Facts	Disorders
	Primary inhibitory NT in CNS. Benzodiazepines, alcohol, barbiturates enhance GABA-A	
Glutamate	**Primary excitatory NT** in CNS. NMDA receptor = LTP = learning	Excess = excitotoxicity (stroke). NMDA hypofunction theory of schizophrenia
Endorphins	Natural opioids. Pain suppression, reward	Opioid addiction

3. Nervous System Organization

```
Nervous System
├── Central (CNS): brain + spinal cord
└── Peripheral (PNS)
    ├── Somatic: voluntary motor + sensory
    └── Autonomic (ANS): involuntary
        ├── Sympathetic: fight-or-flight (NE/ACh, thoracolumbar
T1-L2)
        └── Parasympathetic: rest-and-digest (ACh, craniosacral)
```

Sympathetic activations: ↑ HR, ↑ BP, pupil dilation (mydriasis), bronchodilation, ↓ digestion, ↑ sweat, ↑ glucose. **Parasympathetic:** opposite (except sweat — sympathetic-only via ACh).

4. Brain — Lobes & Functions

Lobe	Primary Functions	Key Areas	Lesion Syndromes
Frontal	Executive, planning, motor, personality	Motor cortex (precentral), Broca's (L inferior frontal, BA44/45), PFC	**Broca's aphasia** (non-fluent), Phineas Gage (ventromedial PFC → personality), contralateral paralysis
Parietal	Somatosensation, spatial, attention	Postcentral gyrus (S1), superior parietal lobule	**Neglect** (R parietal), agraphia, astereognosis, Gerstmann (L angular)
Temporal	Audition, language comp, memory, face	A1 (Heschl's), Wernicke's (L superior temp, BA22), hippocampus,	**Wernicke's aphasia** (fluent, nonsense), prosopagnosia,

Lobe	Primary Functions	Key Areas	Lesion Syndromes
		amygdala, fusiform	anterograde amnesia (H.M.)
Occipital	Vision	V1 (primary visual cortex), BA17	Cortical blindness, achromatopsia, blindsight

Key subcortical structures: - **Thalamus:** sensory relay (ALL senses except olfaction go through it) - **Hypothalamus:** homeostasis. Lateral = hunger; Ventromedial (VMH) = satiety; SCN = circadian; Preoptic = thermoregulation - **Basal ganglia:** motor control, habit. Includes caudate, putamen, globus pallidus, substantia nigra, subthalamic. Damaged in Parkinson's, Huntington's - **Limbic system:** emotion + memory. **Papez circuit:** hippocampus → fornix → mammillary → anterior thalamus → cingulate → hippocampus. Includes amygdala (fear), hippocampus (memory), hypothalamus - **Cerebellum:** motor coordination, balance, procedural learning. Damage → ataxia, intention tremor, dysmetria - **Brainstem:** midbrain (tectum: superior/inferior colliculi; tegmentum: VTA, substantia nigra), pons, medulla. Medulla controls breathing + HR

5. Sensory Systems

Vision

Path: retina → optic nerve → optic chiasm (nasal fibers cross) → optic tract → LGN of thalamus → V1 (occipital) **Retina:** rods (dim light, periphery, no color, 120M) vs cones (bright, fovea, color, 6M). 3 cone types = trichromatic (S-blue, M-green, L-red). **Trichromatic theory (Young-Helmholtz):** 3 cone types. **Opponent-process theory (Hering):** R-G, B-Y, B-W pairs. Both theories correct at different stages. **Feature detectors (Hubel & Wiesel):** simple, complex, hypercomplex cells in V1. **Ventral stream ("what"):** V1 → temporal (object recognition). **Dorsal stream ("where/how"):** V1 → parietal (spatial, action).

Audition

Path: outer ear → middle ear (ossicles: malleus, incus, stapes) → cochlea (basilar membrane, hair cells in organ of Corti) → auditory nerve → cochlear nuclei → superior olive → inferior colliculus → MGN thalamus → A1 (temporal) **Place theory (Helmholtz):** high freq = base of basilar, low freq = apex. True for high freq. **Frequency theory (Rutherford):** firing rate = frequency. True for low freq (<400 Hz). **Volley principle (Wever):** groups of neurons fire in volleys for mid freq (400-4000 Hz).

Other Senses

- **Gustation:** 5 tastes (sweet, sour, salty, bitter, umami). Taste buds → facial (VII)/glossopharyngeal (IX)/vagus (X) nerves → solitary nucleus → thalamus → gustatory cortex (insula)

- **Olfaction:** olfactory epithelium → olfactory bulb → olfactory tract → piriform cortex. **Only sense that bypasses thalamus.**
- **Somatosensation:** touch, temp, pain, proprioception. Dorsal column (fine touch, proprioception) vs spinothalamic (pain, temp)
- **Vestibular:** inner ear (semicircular canals, otolith organs: utricle + saccule) → vestibular nuclei → cerebellum
- **Kinesthetic:** muscle spindles, Golgi tendon organs → proprioception

6. Endocrine System

Master gland: pituitary. Controlled by hypothalamus. - **Anterior pituitary:** ACTH, TSH, FSH, LH, GH, prolactin - **Posterior pituitary:** ADH (vasopressin), oxytocin (made in hypothalamus, stored here)

HPA axis (stress): Hypothalamus → CRH → Anterior pituitary → ACTH → Adrenal cortex → **cortisol**. Negative feedback to hypothalamus. **Sympathoadrenal:** Adrenal medulla → epinephrine + norepinephrine (fast fight-or-flight). **Thyroid:** T3, T4 → metabolism. Hyper = Graves'; hypo = Hashimoto's, cretinism. **Pineal:** melatonin → circadian rhythm. **Gonads:** testes (testosterone), ovaries (estrogen, progesterone).

7. Motivation — Physiology

Hunger: Lateral hypothalamus (LH) = "feeding center" (lesion → aphagia). VMH = "satiety center" (lesion → hyperphagia, obesity). Hormones: leptin (↓ appetite, from adipose), ghrelin (↑ appetite, from stomach), insulin, CCK. **Thirst:** osmometric (cellular dehydration → lateral preoptic) + volumetric (ECF loss → subfornical organ, angiotensin II). **Sex:** hypothalamus (medial preoptic area in males, VMH in females), androgens/estrogens, SDN (sexually dimorphic nucleus).

8. Emotion & Stress

Theories of emotion: - **James-Lange:** body first, emotion follows ("I'm scared because I'm running") - **Cannon-Bard:** body AND emotion simultaneously (thalamus trigger) - **Schachter-Singer (2-factor):** arousal + cognitive label - **Lazarus:** cognitive appraisal first

Amygdala = fear, emotional memory. **Klüver-Bucy syndrome** (bilateral temporal/amygdala damage): hyperorality, hypersexuality, placidity, visual agnosia. **HPA axis** = chronic stress → cortisol → hippocampal atrophy (memory problems), immune suppression. **GAS (Selye):** Alarm → Resistance → Exhaustion.

9. Learning & Memory

Memory systems: - **Short-term / working:** prefrontal cortex (Baddeley: phonological loop, visuospatial sketchpad, central executive, episodic buffer) - **Long-term explicit (declarative):** hippocampus (encoding), medial temporal lobe. Subtypes: episodic (autobiographical) + semantic (facts) - **Long-term implicit (procedural):** striatum (habits), cerebellum (motor skills)

LTP (long-term potentiation): hippocampal CA1, NMDA receptor-dependent, glutamatergic. Cellular basis of learning. **H.M. (Henry Molaison):** bilateral medial temporal lobectomy → anterograde amnesia, preserved procedural and short-term memory. Proves hippocampus = consolidation, not storage. **Korsakoff's syndrome:** thiamine (B1) deficiency (chronic alcoholism) → anterograde + retrograde amnesia, confabulation. Mammillary bodies + thalamus damage.

10. Sleep & Waking

Stages (NREM 1 → 2 → 3 → REM, ~90-min cycles, 4-6/night): | Stage | EEG | Features | |—|—|—| | Wake (relaxed) | Alpha (8-13 Hz) | Eyes closed, relaxed | | NREM 1 | Theta (4-7 Hz) | Light sleep, transition | | NREM 2 | Sleep spindles, K-complexes | ~50% of sleep | | NREM 3 (SWS) | Delta (<4 Hz) | Deep sleep, GH release, memory consolidation | | REM | Beta-like (fast, low amp) | Dreaming, muscle atonia (pons), rapid eye movements, autonomic variability |

REM rebound after deprivation. **Newborns:** 50% REM; adults: 20-25%. **Circadian:** SCN (suprachiasmatic nucleus of hypothalamus), ~24.2 hr free-running. Entrainment by light via retinohypothalamic tract.

Sleep disorders: - **Insomnia:** difficulty initiating/maintaining sleep - **Narcolepsy:** sudden REM intrusions, cataplexy, hypocretin/orexin deficiency - **Sleep apnea:** obstructive (OSA) vs central - **REM behavior disorder:** loss of REM atonia, act out dreams - **Night terrors/sleepwalking:** NREM3 (SWS) parasomnias

11. Genetics & Behavior

Gene, genotype, phenotype. Dominant vs recessive. Autosomal vs X-linked. **Twin studies:** MZ (identical, 100% genes) vs DZ (fraternal, 50%). Heritability = (rMZ − rDZ) × 2. **Adoption studies:** separate genetic vs environmental. **Down syndrome:** trisomy 21. **Fragile X:** FMR1 gene, #1 inherited cause of intellectual disability. **Huntington's:** autosomal dominant, CAG repeats on chr 4, basal ganglia degeneration. **PKU (phenylketonuria):** autosomal recessive, phenylalanine buildup → IDD if untreated.

12. Neuropsychological Assessment

Major batteries: - **Halstead-Reitan (HRNB):** adult (9-14), child. Categories test, TPT, Speech-sounds, Rhythm, Finger tapping, Trail Making A&B. - **Luria-Nebraska:** 11 scales. More theory-driven (Luria's functional systems). - **NIMHANS Battery** (Indian, MET-favorite): comprehensive, culturally adapted. Includes Rey

AVLT, Complex Figure, Verbal N-back, Design Fluency, Category Fluency, Color Trails, Digit Vigilance, Tower of London. - **AIIMS Battery** (Indian): shorter, clinic-use.

Single tests worth knowing: - **WCST (Wisconsin Card Sorting):** frontal/executive function, perseverative errors - **Trail Making A/B:** attention, set-shifting - **Stroop:** response inhibition, prefrontal - **Rey-Osterrieth Complex Figure:** visuospatial + visual memory - **Rey AVLT:** verbal learning + memory - **Bender-Gestalt:** visuomotor - **MMSE / MoCA:** quick cognitive screens

High-Yield Ones Liners (pure memory)

- Sodium-potassium pump ratio = **3:2**
- Resting potential = **−70 mV**, threshold = **−55 mV**
- Acetylcholine decreased in **Alzheimer's**; dopamine decreased in **Parkinson's**
- Broca = frontal, non-fluent. Wernicke = temporal, fluent-nonsense.
- Thalamus relays ALL senses except **olfaction**
- Phineas Gage = **ventromedial PFC** = personality change
- H.M. = **bilateral hippocampal** lesion → anterograde amnesia
- Korsakoff = **thiamine (B1)** deficiency → confabulation
- SCN = **suprachiasmatic**, master circadian clock
- NREM 3 = **slow-wave sleep = delta = memory consolidation**
- REM = dreaming + **atonia** (except eyes + diaphragm)
- Leptin from **adipose** ↓ appetite; ghrelin from **stomach** ↑ appetite
- Papez circuit: hippocampus → fornix → mammillary → ant. thalamus → cingulate → hippocampus
- Klüver-Bucy = **bilateral temporal/amygdala**, hyperorality + placidity
- Narcolepsy = **hypocretin/orexin** deficiency
- Basal ganglia: caudate + putamen = **striatum**; + globus pallidus = lentiform
- Cerebellum damage → ataxia, **intention tremor**, dysmetria
- Primary visual cortex = **V1 = BA17 = striate cortex**
- NMDA receptor + glutamate = **LTP = learning substrate**

Next Up

- `01-neurons-and-synapses.md` — deeper dive with Pinel chapter refs
- `02-brain-anatomy.md` — lobe-by-lobe with clinical syndromes
- `03-sensory-systems.md` — vision heaviest
- Practice MCQ set (20 Qs matching MET difficulty)

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Part 2: Psychological Assessment

Assessment Primer (Narrative)

Psychological Assessment, Read-First Primer

This is written in explanation mode. Read it like a chapter, not a cheatsheet. Every fact has a "why" attached. Once this clicks, the existing flashcard decks (Assessment 1 and 2) become useful as a self-test, not a confusion machine.

If you walk away with one mental model, it is this: **a psychological test is a measuring instrument, and like any instrument, it has to be (a) consistent and (b) accurate.** Consistent = reliability. Accurate = validity. Everything else in this section is either a way to estimate reliability, a way to estimate validity, a way to interpret scores against norms, or an example of a real-world test. That is the whole map.

1. Why test construction is even a thing

Psychology is uncomfortably stuck between two facts. First, the things we want to measure (anxiety, intelligence, personality) are not directly observable. There is no anxiety-meter you can clip to a fingertip. Second, we still need to make decisions: should this child be in special education? Is this candidate fit for the army? Does this patient have OCD? So we build instruments out of items, like a thermometer made of questions.

That is why item analysis exists. Each item is a single sensor reading. If the sensor is too easy (everyone gets it right) it tells us nothing. If the sensor is too hard (no one gets it) it tells us nothing. If the sensor agrees with the rest of the test, it is doing its job. If it disagrees, something is wrong with that item.

Difficulty index, p

The difficulty index is just the proportion who got an item right.

```
p = (number who got the item right) / (total who attempted)
```

Range zero to one. $p = 0.50$ means half the test-takers got it right. That is the sweet spot for a free-response item, because it discriminates the most. For a 4-option MCQ, the sweet spot is closer to 0.625 because guessers will pick the right answer 25% of the time by chance, and you want item difficulty to land halfway between guessing (0.25) and ceiling (1.00).

Items with p above 0.80 are too easy and items below 0.20 are too hard. Both are candidates for revision because they cannot tell strong students from weak ones.

Discrimination index, D

This one asks: do the test-takers who scored well on the whole test do better on this item than the test-takers who scored badly? Specifically:

```
D = p_upper − p_lower
```

Where p_upper is the proportion correct in the top 27% of scorers and p_lower is the proportion in the bottom 27%. Range is minus one to plus one.

If D is high (above 0.40), the item agrees with the overall test. Top scorers got it right, bottom scorers got it wrong. The item is a good signal.

If D is near zero, the item is random noise. Both groups perform similarly.

If D is **negative**, something is wrong. The bottom group out-performed the top group. That happens when an item is mis-keyed, ambiguously worded, or has a trick that strong test-takers over-think. Negative D means discard immediately.

Always read negative D as a red flag, not as a hard item.

2. Reliability, the consistency story

Reliability is not about whether the test is correct. It is about whether the test is **consistent**. A bathroom scale that reads 75 kg every time you step on it is reliable, even if it is calibrated wrong and the true weight is 70. A scale that bounces between 60, 90, and 80 is unreliable, full stop.

Why does this matter? Because if a test is unreliable, you cannot trust any single score from it. The next time you give the same test to the same person, you will get a different answer for reasons that have nothing to do with the trait you are trying to measure. You are measuring noise.

The classical test theory equation summarizes this:

```
Observed Score = True Score + Error
```

Reliability is the proportion of observed score variance that is true variance, not error. A reliability of 0.80 says 80% of what you are seeing is signal, 20% is noise.

There are several reliability methods because there are several ways to think about “consistency”:

- **Across time:** test the same person twice on different days (test-retest).
- **Across forms:** give two different versions of the test (parallel forms).
- **Within one administration, across items:** check that the items hang together (split-half, Cronbach alpha, KR-20).
- **Across raters:** check that two judges agree (Cohen kappa for categorical, ICC for continuous).

Test-retest, the time-stability check

Give the test on Monday, give it again three weeks later, correlate the two scores. If the correlation is high, the test is stable across time.

The trap: if you wait too long, the trait itself can change, and the correlation drops not because the test is bad but because the person actually changed. If you wait too short, memory carryover inflates the correlation. The standard window is two to four weeks.

Split-half plus Spearman-Brown

Give the test once, split the items into two halves (often odd-numbered vs even-numbered), and correlate the two halves. The result tells you how internally consistent the test is.

Catch: by splitting, you have effectively halved the test length, and shorter tests are less reliable. So the raw split-half correlation underestimates the full-test reliability. Spearman-Brown corrects for that:

```
r_full = 2 * r_half / (1 + r_half)
```

If r_half is 0.60, the full-test estimated reliability is 1.20 / 1.60 = 0.75. Spearman-Brown is the standard fix you apply after any split.

Cronbach alpha

Imagine doing every possible split-half (odd-even, first-half last-half, items 1-3-5 vs 2-4-6, etc.) and averaging the result. That mean is Cronbach alpha. It is the most common internal-consistency measure for tests with multi-point items (like Likert).

Important: alpha is **not** a measure of unidimensionality. A test can have high alpha and still be measuring two distinct constructs. Alpha rises with item count, item homogeneity, and variance in the trait. The MET love this trap.

KR-20

When items are right-or-wrong (dichotomous, 1 or 0), Cronbach alpha simplifies to a special formula called KR-20 (Kuder-Richardson 20). It is the same idea, just adapted for 0/1 items. So if you see “30-item right-wrong vocabulary test, one administration, which reliability?”, the answer is KR-20.

Inter-rater, kappa and ICC

What if humans do the scoring (Rorschach protocols, clinical interviews, essay grades)? Then reliability is about whether two raters agree. For categorical decisions (does the patient have OCD, yes or no?), Cohen kappa corrects for chance agreement. Kappa = 0.81 or above is almost-perfect; 0.61 to 0.80 is substantial.

For continuous ratings (a 0-100 severity score), use the intraclass correlation coefficient (ICC). Different forms of ICC exist depending on whether raters are random or fixed.

Critical: kappa is **not** an internal-consistency measure. If a question describes a one-administration internal-structure test, kappa is wrong.

Standard Error of Measurement (SEM)

SEM is the practical takeaway of reliability. It tells you the band of error around any one observed score.

```
SEM = SD * sqrt(1 - r_xx)
```

where SD is the test's standard deviation and r_xx is its reliability.

Example: a test with SD = 20 and reliability 0.81 has SEM = 20 * sqrt(0.19) = 20 * 0.436 = 8.72. So if a person scores 100, the 68% confidence band is roughly 91 to 109 (one SEM either side).

Higher reliability shrinks SEM. SEM is what lets you say "this score is meaningfully different from that one" or "this is within the noise band."

3. Validity, the accuracy story

A test can be perfectly reliable and still measure the wrong thing. Validity asks whether the test measures the construct it claims to measure.

There are three broad evidence types in the classical model. (Modern theorists, especially Messick, fold them all into "construct validity," but MET still uses the classical breakdown so know both.)

Content validity

Does the test cover the domain it claims to cover? An anxiety scale that only asks about racing heart but ignores worry, restlessness, and sleep is content-deficient. Content validity is established by expert panels who rate whether items adequately sample the content domain. Lawshe's CVR is the technical formula but expert judgment is the core.

Do not confuse with face validity. Face validity is whether the test **looks** valid to a layperson taking it. That is the weakest form because laypeople are not domain experts. A test can have high face validity and bad content validity (looks plausible but misses key content) or low face validity and high content validity (rigorous content but feels weird to take).

Criterion validity

Does the test correlate with a real-world outcome (the criterion)? Two flavors:

- **Concurrent**: criterion measured at the same time. Example: a new depression scale today vs an established gold standard like HAM-D today. Used when a gold standard already exists.
- **Predictive**: criterion measured later. Example: GRE today predicts graduate GPA two years later. Used when you want to forecast future behavior.

Memory hook: **Predictive = Postponed.** Concurrent = Current.

Construct validity

The most abstract form. It asks: if my test really measures Construct X, then it should correlate strongly with other measures of X (convergent), correlate weakly with measures of unrelated constructs (discriminant), and behave the way theory predicts in study after study (nomological network).

The classic technique here is **Campbell and Fiske's 1959 multitrait-multimethod matrix (MTMM)**, which tests convergent and discriminant validity simultaneously by measuring multiple traits with multiple methods.

Validity ceiling

Validity cannot exceed the square root of reliability:

```
Maximum validity coefficient between Test A and Test B =
sqrt(r_xx_A * r_xx_B)
```

If A is 0.64 reliable and B is 0.81 reliable, the maximum correlation between them is sqrt(0.64 * 0.81) = 0.72. Unreliable tests cannot be strongly valid. Reliability is necessary for validity, not sufficient.

4. Norms and standard scores, the comparison story

A raw score by itself means nothing. "Sahil scored 47 on a vocabulary test" tells you nothing until you know what other people score. Norms convert raw scores into a frame of reference.

The two most useful conversions are percentile ranks and standard scores.

Percentile ranks

The percentage of the norm group scoring at or below the person's score. A 75th percentile means 75% of people scored at or below this person. Easy to interpret, but ordinal (the gap between the 50th and 60th percentile is much smaller than the gap between the 90th and 99th).

Standard scores

These convert raw scores onto an interval scale where the mean and SD are fixed. The conversion is always:

```
Standard Score = M + (SD * z)
```

where z is the person's z-score from the norm group.

You need to memorize a handful of standard score systems cold. They get tested every cycle:

Score	Mean	SD	Use
z	0	1	Universal
T	50	10	MMPI, Big Five inventories
Wechsler IQ	**100**	**15**	WAIS, WISC, WPPSI
Old Stanford-Binet IQ	100	16	Older SB versions
Stanine	5	2	Achievement tests, range 1 to 9
Sten	5.5	2	16PF
Wechsler subtest scaled	10	3	Subtests inside WAIS/WISC, range 1 to 19

The most-tested traps: - Wechsler IQ uses SD = 15. Old Binet used SD = 16. Read the question carefully. - Stanine has SD = 2, **not** 1. Common distractor. - T is 50/10, **not** 100/10.

Normal-curve landmarks worth memorizing: 68% within 1 SD, 95% within 2 SD, 99.7% within 3 SD. A z of plus 1 lands at the 84th percentile. A z of plus 2 lands at the 98th percentile.

5. The major intelligence tests

The Wechsler family

Wechsler is the workhorse. Three versions for three age bands:

- **WPPSI** (Preschool and Primary), ages 2:6 to 7:7, currently version IV.
- **WISC** (Children), ages 6:0 to 16:11, currently version V.
- **WAIS** (Adult), ages 16:0 to 90:11, currently version IV (with WAIS-5 in rollout).

WAIS-IV produces a Full-Scale IQ (FSIQ) plus four index scores: Verbal Comprehension, Perceptual Reasoning, Working Memory, Processing Speed. WISC-V expanded to five indices (Verbal Comprehension, Visual Spatial, Fluid Reasoning, Working Memory, Processing Speed) by separating fluid reasoning from visual-spatial.

Stanford-Binet

The granddaddy. Started as Binet-Simon in France in 1905, brought to Stanford by Lewis Terman in 1916 with the famous IQ ratio formula (IQ = MA / CA times 100). Modern SB-5 abandoned the ratio in favor of deviation IQ (M = 100, SD = 15 since 2003).

Raven's Progressive Matrices

Nonverbal, pattern-completion, culture-reduced. Used widely in cross-cultural research and India. Three forms: Standard (SPM), Coloured (CPM, for children and elderly), Advanced (APM, for above-average ability).

Indian adaptations (high yield for both MET and NIMHANS)

- **Bhatia Battery of Performance Tests of Intelligence**: C.M. Bhatia, 1955. **Five subtests**: Koh's Block Design, Alexander's Pass-Along, Pattern Drawing, Immediate Memory, Picture Construction. For ages 11 to 16 with Indian norms.
- **Malin's Intelligence Scale for Indian Children (MISIC)**: WISC adaptation, ages 6 to 15.
- **Binet-Kamat Test (BKT)**: Kamat's Marathi/Kannada adaptation of the Binet.
- **Seguin Form Board**: Nonverbal, ages 3 to 7, measures speed of visual-motor coordination.

Bhatia having five subtests is a chronic MCQ trap. Memorize the count.

6. Personality assessment, two big families

Objective (self-report)

The big four to know:

- **MMPI-2 / MMPI-2-RF / MMPI-3**: 10 clinical scales (Hs, D, Hy, Pd, Mf, Pa, Pt, Sc, Ma, Si) plus validity scales (L, F, K). Authored by Hathaway and McKinley. 567 true-false items in MMPI-2. **The number is 10 clinical, not 8.**
- **NEO-PI-3**: Big Five (OCEAN: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism). 240 items. By Costa and McCrae.
- **16PF**: Cattell's 16 primary factors plus 5 globals. 185 items. **16PF is not Big Five**; they are different theories.
- **EPQ-R**: Eysenck's three big traits (Extraversion, Neuroticism, Psychoticism) plus Lie scale.

Projective (ambiguous stimulus)

Theory: ambiguous stimuli pull unconscious material to the surface (projection).

- **Rorschach**: **10 inkblots**, 5 chromatic and 5 achromatic. Hermann Rorschach, 1921. Modern scoring via Exner Comprehensive System or R-PAS.
- **TAT (Thematic Apperception Test)**: **31 cards total**, 20 administered per session selected by age and sex of subject. Henry Murray and Christiana Morgan, 1935. Theory: needs-press.
- **CAT (Children's Apperception Test)**: Bellak, 1949. Animal cards for children.
- **Sentence Completion (RISB)**: Rotter.
- **Draw-a-Person, House-Tree-Person, Bender-Gestalt**: drawing-based; HTP and DAP are Buck/Machover.

The card-count traps the MET asks every cycle: - Rorschach = **10 cards** (not 12, not 15, not 20). - TAT = **31 total** (not 10, not 20).

7. Aptitude, attitude, interest, briefly

- **DAT (Differential Aptitude Test)**: 8 subtests (Verbal, Numerical, Abstract, Mechanical, Space, Spelling, Language, Perceptual). Bennett, Seashore, Wesman.
- **GATB**: nine aptitudes, US Employment Service.

Attitude scales: - **Thurstone (1928)**: equal-appearing intervals, judges scale items 1 to 11. - **Likert (1932)**: summated rating, typically 5-point. - **Guttman (1944)**: cumulative; endorsing a stronger item implies endorsing weaker ones. - **Bogardus**: social distance. - **Osgood semantic differential**: bipolar adjectives, 3 factors (evaluation, potency, activity).

Interest inventories: - **Strong Interest Inventory**: Holland themes (RIASEC) plus occupational scales. - **Kuder Career Search**: forced-choice triads. - **Holland Self-Directed Search**: RIASEC (Realistic, Investigative, Artistic, Social, Enterprising, Conventional).

8. Psychiatric rating scales, the clinical-tool family

These come up constantly. Memorize what each scale measures and whether it is clinician-rated or self-rated.

Scale	Measures	Type
BDI-II	Depression	Self, 21 items
HAM-D / HDRS	Depression	Clinician, 17-21 items
MADRS	Depression	Clinician, 10 items
PHQ-9	Depression	Self, 9 items
BAI	Anxiety	Self, 21 items
HAM-A	Anxiety	Clinician, 14 items
GAD-7	Anxiety	Self, 7 items

Scale	Measures	Type
Y-BOCS	**OCD severity (gold standard)**	Clinician, 10 items
PANSS	**Schizophrenia symptoms (positive + negative)**	Clinician, 30 items
BPRS	Brief psychiatric symptoms	Clinician, 18 items
YMRS	Mania	Clinician, 11 items
MMSE	**Cognitive screening (general)**	30 points, cutoff 23-24
MoCA	**Mild cognitive impairment**	30 points, cutoff 25, more sensitive than MMSE
C-SSRS	Suicide severity	—
CAPS-5	PTSD	Clinician
AUDIT	Alcohol use	Self

The chronic confusions: - Y-BOCS is **OCD specifically**, not anxiety in general. If a question mentions "obsessive-compulsive severity gold standard", the answer is Y-BOCS. - PANSS is **schizophrenia**, covers both positive and negative symptoms. BPRS is briefer and more general. - MMSE has **30 points** with a cutoff in the 23-24 range. Frequently asked.

9. Educational measurement basics

Three distinctions worth knowing:

- **Achievement** measures what you have learned (past). Examples: WIAT, school finals.
- **Aptitude** measures what you can learn (future). Examples: DAT, SAT.
- **Intelligence** measures general cognitive capacity. Examples: WAIS, SB.

Scoring frame: - **Norm-referenced**: compare student to peer group (ranking, selection). - **Criterion-referenced**: compare student to absolute standard (mastery, licensing).

Timing: - **Formative**: during learning, low-stakes feedback. - **Summative**: end of unit, high-stakes outcome.

10. The 10 traps to memorize cold

1. Validity has a ceiling at sqrt(reliability). Unreliable tests cannot be strongly valid.
2. Negative discrimination index = miskeyed item. Discard immediately.
3. Cronbach alpha is not unidimensionality. High alpha can mask multiple factors.
4. Wechsler IQ uses SD 15. Old Stanford-Binet used SD 16.
5. Stanine has SD 2, range 1 to 9.
6. Predictive = future criterion. Concurrent = same-time criterion.
7. Face validity is the weakest form. Content validity is expert-judgment of domain coverage.

8. MMPI-2 has 10 clinical scales (Hs D Hy Pd Mf Pa Pt Sc Ma Si).
9. Rorschach = 10 inkblots. TAT = 31 total cards (20 per admin).
10. Y-BOCS = OCD. PANSS = schizophrenia. MMSE = 30 points, cutoff 23-24.

11. How to use this primer with the flashcard app

Read this top to bottom, even the sections you think you know. Then go to:

https://msahil515.github.io/sihan-met-flashcards/

Open Psych Assessment Deck 1 (Reliability, Validity, Norms). Tap each card. **Try to answer in your head before flipping.** That step is the whole point. Mark Knew or Missed honestly. Then Deck 2 (Tests and Scales).

After both decks, you will know roughly which 3-5 facts are not sticking. Tell me which Q numbers you Missed and I will hammer those specifically next session.

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Reliability and Validity (Formulas)

Reliability & Validity — Deep Dive

Single heaviest MCQ topic in Psychological Assessment. Memorize every row.

Reliability (consistency)

Conceptual core

Observed score = True score + Error. Reliability coefficient = ratio of true variance to observed variance. $r_{xx} = \sigma^2_{true} / \sigma^2_{observed}$

A score of $r_{xx} = 0.80$ means **80% of variance in observed scores reflects true individual differences**, 20% is measurement error.

All types in one table

Method	What error source it estimates	Procedure	Coefficient	MCQ trap
Test-retest	Time stability	Same test administered twice, typically 2+ wks apart	Pearson r (test-retest r)	Too short interval → memory carry-over in-

Method	What error source it estimates	Procedure	Coefficient	MCQ trap
				flates r; too long → real change in trait lowers r. "Carry-over effect" is the classic answer.
Alternate forms / Parallel forms	Content sampling + time (if delayed)	Two "equivalent" versions of the test given to same subjects	Pearson r	Only works if forms are truly parallel (same M, SD, item content). Hard to build.
Split-half	Content sampling (one sitting)	Split items odd/even; correlate halves; apply Spearman-Brown correction	Spearman-Brown corrected r	The raw half-to-half correlation **underestimates** full-test reliability because the test is now half as long. Spearman-Brown "uncorrects" this.
Cronbach's α (Coefficient alpha)	Internal consistency (all items)	Computed in one pass over all	α	Not a measure of unidimen-

Method	What error source it estimates	Procedure	Coefficient	MCQ trap
		items; mathematically equal to mean of all possible split-halves		sionality. A test with two uncorrelated subscales can still have high α if items are many and inter-related.
KR-20	Internal consistency, dichotomous items (right/wrong, yes/no)	Kuder-Richardson formula 20	α-equivalent	KR-20 is **a special case of α** when items are dichotomous. KR-21 assumes equal item difficulty (simpler but less accurate).
Inter-rater (categorical)	Rater subjectivity	Two raters score same protocols	Cohen's κ (kappa)	κ corrects for chance agreement; κ = 1 perfect, κ = 0 chance, κ < 0 worse than chance. 0.81–

Method	What error source it estimates	Procedure	Coefficient	MCQ trap
				1.0 almost perfect; 0.61–0.80 substantial.
Inter-rater (continuous)	Rater subjectivity, ratio data	Two or more raters on ratings/ scores	Intraclass Correlation Coefficient (ICC)	ICC has multiple forms (1,1 vs 2,1 vs 3,1 — Shrout & Fleiss). Random vs fixed raters.

Key formulas (MCQ cares)

- **Spearman-Brown:** $r_{full} = (n \cdot r_{half}) / (1 + (n-1) \cdot r_{half})$. For split-half, $n = 2$.
- **Standard Error of Measurement (SEM):** $SEM = SD_{test} \cdot \sqrt{(1 - r_{xx})}$. Confidence band around observed score.
- **Reliability and test length:** lengthening a test increases reliability (by Spearman-Brown), approaching but never reaching 1.0.

Factors that raise α

1. More items (reliably measuring same construct).
2. Homogeneity of items (higher inter-item correlations).
3. Greater sample variability in the trait (restricted range lowers α).

Thresholds by use

- ≥ 0.95: medical/legal decisions on individuals.
- 0.90: high-stakes selection, clinical diagnosis.
- 0.80: research on individuals.
- 0.70: research on groups, early-stage scales.
- < 0.70: unacceptable for individual decisions.

Validity (accuracy)

Messick's unified view (1989, DSM-era MCQs)

Construct validity is the **single unifying concept**; content, criterion, and face validity are sources of evidence, not separate validities. MET still asks the classical three-part schema, so know both.

Classical three-part schema

1. Content validity

- **Question:** Do items adequately sample the target domain?
- **Evidence:** Expert panel, table of specifications, Lawshe's CVR (Content Validity Ratio = $(n_e - N/2) / (N/2)$, where n_e = experts rating item essential).
- **Face validity** ≠ content validity. Face = looks valid to laypeople; content = judged by subject experts against domain specifications.

2. Criterion-related validity

- **Question:** Does the test correlate with a gold-standard outcome?
- **Concurrent:** test and criterion measured at same time (e.g., new depression scale vs HAM-D today).
- **Predictive:** test now, criterion later (e.g., GRE → grad GPA 2 years out).
- **Evidence:** validity coefficient (r between test and criterion). 0.30–0.40 is typical; > 0.50 is excellent.

Criterion contamination: when the criterion score is influenced by the predictor. E.g., if a counselor knows a client's admission test score and uses it to grade the client later, the "predictive validity" is inflated.

3. Construct validity

Most abstract. Evidence accumulates over many studies. - **Convergent:** high correlation with measures of related constructs. - **Discriminant:** low correlation with measures of unrelated constructs. - **Factorial:** factor-analytic structure matches theory. - **Nomological network:** test scores relate to other variables in ways theory predicts.

Other forms

- **Incremental validity:** does this test add predictive power *beyond* existing predictors? Evidenced by ΔR^2 in hierarchical regression.
- **Ecological validity:** do test results generalize to real-world functioning?
- **Consequential validity:** Messick — what are the social consequences of test use? (Bias, adverse impact.)

Campbell & Fiske (1959) — Multitrait-Multimethod Matrix (MTMM)

Gold-standard design for establishing convergent + discriminant validity simultaneously. - Measure multiple traits via multiple methods. - Same trait, different methods (monotrait-heteromethod) = convergent. Should be **high**. - Different trait, same method (heterotrait-monomethod) = method variance. Should be **low** (but usually isn't, hence "method bias"). - Different trait, different method = heterotrait-heteromethod. Should be lowest.

Validity coefficients and corrections

- **Attenuation correction:** r_xy_true = r_xy_observed / √(r_xx · r_yy). Estimates what the correlation would be if both measures were perfectly reliable.
- **Validity cannot exceed √reliability.** Key implication for exam: an unreliable predictor cannot be strongly valid.

Reliability–Validity Relationship Cheat

	Reliable	Unreliable
Valid	Good test	Impossible
Invalid	Consistent but wrong — possible (e.g., biased scale)	Useless

A bathroom scale 5 kg off every time is reliable (consistent) but invalid (wrong number). A scale giving random numbers is neither. A scale reading correctly is both. Reliability is necessary but not sufficient for validity.

Drilled MCQ Points

1. Which reliability coefficient for a dichotomously-scored 50-item vocabulary test, one administration? → **KR-20.**
2. Which validity for predicting first-year grad GPA from GRE? → **Predictive (criterion).**
3. New anxiety scale correlates 0.75 with STAI (given concurrently). What does this support? → **Convergent validity** (subtype of construct).
4. r_xx = 0.80, SD = 10. What is SEM? → 10 · √0.20 = **4.47.**
5. If a 20-item test has r_xx = 0.60, what is the estimated reliability at 40 items? → n = 2, Spearman-Brown: (2 · 0.60) / (1 + 0.60) = **0.75.**
6. A test shows D = −0.25 for an item. Interpretation? → **Miskeyed or misleading; discard.**
7. What is the maximum theoretical correlation between Test X (r_xx = 0.64) and Test Y (r_yy = 0.81)? → √(0.64 · 0.81) = **0.72.**
8. A test looks like it measures depression but was never validated against DSM criteria. This is ___ but not ___. → **Face valid; content/criterion valid.**
9. Which scale has inter-rater coefficient most relevant? → **Rorschach, projectives, HAM-D, PANSS** — clinician-scored.

10. Cronbach's α of 0.92 for a 40-item scale drops to 0.68 when items are halved. Why? → **Test length directly affects α** (Spearman-Brown logic applies to α as well as split-half).

Memorization Hooks

- **CONVERGENT = CONNECT.** Similar things should correlate.
- **DISCRIMINANT = DISTINCT.** Dissimilar things shouldn't.
- **"α is not unidimensionality"** — if a question says "high α means test measures one construct", it's wrong.
- **Reliability ceiling mnemonic:** "A valid test must be reliable, but a reliable test may be invalid." Necessary ≠ sufficient.
- **Predictive vs concurrent: P**redictive = **P**ostponed. Concurrent = same time.

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Assessment Master Cheatsheet

Psychological Assessment — Master Cheatsheet

High-yield table for MET 2026. Compress the whole section into 5 pages. Drill these until automatic.

1. Test Construction — Item Analysis

Difficulty Index (p)

- **Formula:** p = (number who got item right) / (total who attempted)
- **Range:** 0.00 (no one got it) → 1.00 (everyone got it)
- **Ideal for MCQ:** p ≈ 0.50 (with guessing correction, target 0.5 + (0.5 / k) where k = options; for 4-option MCQ, target ≈ 0.625)
- **Easy item:** p > 0.80. **Hard item:** p < 0.20. Both have low discrimination; usually cut.

Discrimination Index (D)

- **Formula:** D = p_upper − p_lower, using top 27% and bottom 27% scorers.
- **Range:** −1.00 to +1.00.
- **Ebel's rule:**
  - D ≥ 0.40 = excellent, keep.
  - 0.30–0.39 = good, keep.
  - 0.20–0.29 = marginal, revise.
  - < 0.20 = poor, reject or rewrite.

    ◦ Negative D = lower group outperformed upper → item is **miskeyed or misleading**. Instant reject.

Distractor Analysis

- Every wrong option (distractor) should attract some low scorers.
- A distractor chosen by **zero** people = not functioning, replace.
- A distractor chosen more by high scorers than low = it's plausible or ambiguous, revise.

2. Reliability (consistency, not accuracy)

Type	What it checks	How	Coefficient
Test-retest	Stability across time	Same test, same people, 2+ weeks apart	Pearson r
Parallel forms	Equivalence of two versions	Two comparable versions, same people	Pearson r
Split-half	Internal consistency (quick)	Score odd vs even items, correct with Spearman-Brown	Spearman-Brown
Cronbach's α	Internal consistency (all items)	Computed over all items simultaneously	α, 0–1
KR-20 / KR-21	Cronbach's α for dichotomous (right/wrong) items	Kuder-Richardson formula	α equivalent
Inter-rater	Agreement between scorers	2 raters score same	κ (Cohen's kappa) or ICC

Type	What it checks	How	Coefficient
		protocols	

Thresholds: - α ≥ 0.90: excellent, required for high-stakes tests. - α 0.80–0.89: good, acceptable for research. - α 0.70–0.79: acceptable for early-stage scales. - α < 0.70: poor, scale needs revision.

MCQ traps: - Cronbach's α **increases** when: more items, items more homogeneous, higher sample variance. - α is **not** a measure of unidimensionality. High α ≠ single construct. - Spearman-Brown corrects split-half because halving a test lowers reliability; the formula estimates what reliability would be at full length.

3. Validity (accuracy — does it measure what it claims)

Type	Subtype	What it asks	How evidenced
Content	—	Do items cover the domain?	Expert panel, table of specifications
Face	—	Does it *look* like it measures X?	Subjective; weakest form; good for motivation, not science
Criterion	**Concurrent**	Does it correlate with a current gold standard?	Correlate with established test taken at same time
Criterion	**Predictive**	Does it forecast future behavior?	Correlate with outcome measured later (e.g., GRE → grad GPA)
Construct	**Convergent**	Does it correlate with similar constructs?	High r with conceptually related measures
Construct	**Discriminant**	Is it *distinct* from unrelated constructs?	Low r with conceptually unrelated measures
Construct	**Factorial**	Do items load as expected on factors?	Factor analysis (EFA, CFA)
Incremental	—	Does it add predictive power over existing tests?	ΔR^2 in regression
Ecological	—	Do results generalize to real-world settings?	Field validation

Campbell & Fiske (1959) MTMM matrix: classic technique to establish convergent + discriminant validity simultaneously. High correlations for same trait across methods (convergent) + low correlations for different traits within same method (discriminant).

Reliability ceiling: validity cannot exceed √reliability. An unreliable test cannot be valid. Reliable test can still be invalid.

4. Norms & Standard Scores

Types of Norms

- **Age norms:** mental age (MA), developmental quotient (DQ). Used in Binet, Bayley.
- **Grade norms:** achievement tests. “Grade equivalent 7.3” = average 7th-grader in 3rd month.
- **Percentile ranks:** % of norm group scoring at or below. Ordinal, not interval. The middle ranks (25–75) are crowded; extremes are spread out.
- **Standard scores (interval):** z, T, stanine, IQ.

Standard Score Conversions (memorize)

Score	Mean	SD	Range	Example
z	0	1	typically −3 to +3	z = (X − M) / SD
T	50	10	20–80	MMPI, Big Five
IQ (Wechsler)	100	15	~55–145	WAIS, WISC
IQ (Stanford-Binet old)	100	16	~52–148	older SB editions
Stanine	5	2	1–9	achievement tests
Sten	5.5	2	1–10	16PF
Scaled (Wechsler subtests)	10	3	1–19	WAIS/WISC subtests
SAT/GRE section	500 (old) / 150 (new GRE)	100 / 10	200–800 / 130–170	—
Normal Curve Equivalent (NCE)	50	21.06	1–99	education research

Conversion: any standard score = M + (SD × z). E.g., T = 50 + 10z. IQ = 100 + 15z.

Normal Curve Landmarks

- ±1 SD = 68.26%
- ±2 SD = 95.44%
- ±3 SD = 99.74%
- Percentiles at key z: z=−1 → 16th, z=0 → 50th, z=+1 → 84th, z=+2 → 98th.

5. Intelligence Tests

Wechsler Family

Test	Age range	Latest (as of 2026)
WPPSI (Preschool)	2:6 – 7:7	WPPSI-IV (2012)
WISC (Children)	6:0 – 16:11	WISC-V (2014)
WAIS (Adult)	16:0 – 90:11	WAIS-IV (2008); WAIS-5 releasing

WAIS-IV structure: 4 indices → FSIQ - Verbal Comprehension (VCI): Similarities, Vocabulary, Information, (Comprehension) - Perceptual Reasoning (PRI): Block Design, Matrix Reasoning, Visual Puzzles, (Picture Completion, Figure Weights) - Working Memory (WMI): Digit Span, Arithmetic, (Letter-Number Sequencing) - Processing Speed (PSI): Symbol Search, Coding, (Cancellation)

WISC-V moved to 5 indices: Verbal Comprehension, Visual Spatial, Fluid Reasoning, Working Memory, Processing Speed.

Stanford-Binet

- Originally Binet-Simon (1905), revised by Terman at Stanford (1916).
- Introduced **IQ = (MA/CA) × 100** (ratio IQ, obsolete).
- Modern SB-5 uses deviation IQ (M=100, SD=15 after 2003 revision; older was SD=16).
- Five factors: Fluid Reasoning, Knowledge, Quantitative, Visual-Spatial, Working Memory.

Raven's Progressive Matrices

- Nonverbal, culture-reduced, measures **g (fluid intelligence)**.
- Three forms: Standard (SPM), Coloured (CPM, children/elderly), Advanced (APM, high ability).
- Key for cross-cultural and Indian contexts.

Kaufman (KABC-II, KAIT)

- Emphasizes process (sequential vs simultaneous) per Luria.

Indian Adaptations (MCQ favorites)

- **Bhatia Battery of Performance Tests of Intelligence** (C.M. Bhatia, 1955): 5 subtests — Koh's Block Design, Alexander's Pass-Along, Pattern Drawing, Immediate Memory, Picture Construction. For 11–16 yrs, Indian norms.
- **Malin's Intelligence Scale for Indian Children (MISIC):** WISC adaptation, 6–15 yrs.
- **Binet-Kamat Test (BKT):** Kamat's Kannada/Marathi adaptation of Binet.
- **Seguin Form Board:** nonverbal, 3–7 yrs, measures speed of visual-motor.
- **CPM (Coloured Progressive Matrices):** widely used in India for 5–11 yrs.

6. Personality Assessment

Objective (self-report)

Test	Constructs	Items	Notes
MMPI-2	10 clinical scales + validity	567 T/F	Hs, D, Hy, Pd, Mf, Pa, Pt, Sc, Ma, Si
MMPI-2-RF	Restructured clinical (RC) scales	338	2008 revision, more parsimonious
MMPI-3	Latest (2020)	335	—
NEO-PI-3	Big Five (OCEAN) + 30 facets	240	Costa & McCrae
NEO-FFI-3	Big Five only	60	short form
16PF	16 primary factors + 5 globals	185	Cattell
EPQ-R	Extraversion, Neuroticism, Psychoticism, Lie	100	Eysenck
CPI	20 folk-concept scales	434	Gough; "normal" populations

MMPI validity scales (high yield): - L (Lie): naive faking good - F (Infrequency): random, exaggerating symptoms - K (Correction): subtle defensiveness - ?, VRIN, TRIN (inconsistency indicators)

Big Five mnemonic: OCEAN — Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism.

Projective

Assumption: ambiguous stimuli pull out unconscious material (projection).

Test	Stimulus	Primary author	Scoring
Rorschach	10 inkblots (5 chromatic, 5 achromatic)	Hermann Rorschach, 1921	Exner Comprehensive System; R-PAS (2011)
TAT	31 ambiguous picture cards (20 per admin by age/sex)	Murray & Morgan, 1935	Murray's needs-press; SCORS
CAT	Animal cards for children	Bellak, 1949	Child adaptation of TAT
SCT / Sentence	"I wish…" incomplete sentences	Rotter (RISB)	Various

Test	Stimulus	Primary author	Scoring
Completion			
Draw-a-Person (DAP)	Draw a person	Machover, 1949; Goodenough-Harris	Scored for body image, cognitive maturity
House-Tree-Person (HTP)	Draw house, tree, person	Buck, 1948	Personality + intelligence screen
Bender-Gestalt	Copy 9 geometric figures	Bender, 1938	Visual-motor, organic screen

Rorschach Exner codes (common in MCQs): - Location: W (whole), D (common detail), Dd (unusual detail), S (space) - Determinant: F (form), M (human movement), FC (form-color), C (pure color), Y (shading/diffuse) - Content: H (human), A (animal), An (anatomy) - Popular: P (common response)

7. Aptitude Tests

- **DAT (Differential Aptitude Test):** 8 subtests — Verbal, Numerical, Abstract, Perceptual, Mechanical, Space, Spelling, Language. Bennett, Seashore & Wesman.
- **GATB (General Aptitude Test Battery):** 9 aptitudes. US Employment Service.
- **AAT (Army Alpha & Army Beta):** WWI; Alpha verbal, Beta nonverbal.
- **Seashore Measures of Musical Talent.**
- **Bennett Mechanical Comprehension Test.**

8. Attitude Scales

Method	Year	Key idea
Thurstone	1928	Equal-appearing intervals; judges rate items 1–11; item scale values assigned
Likert	1932	Summated rating, 5-point (SA-A-N-D-SD); simplest to build
Guttman	1944	Cumulative/unidimensional; endorsing stronger item implies weaker ones
Bogardus Social Distance	1925	7 steps of accepted intimacy with out-group
Osgood Semantic Differential	1957	Bipolar adjectives; 3 factors: evaluation, potency, activity

9. Interest Inventories

- **Strong Interest Inventory (SII):** Holland themes (RIASEC) + basic interests + occupational scales.
- **Kuder Career Search/Preference:** forced-choice triads; 10 interest areas.
- **Holland's Self-Directed Search (SDS):** RIASEC (Realistic, Investigative, Artistic, Social, Enterprising, Conventional).
- **Campbell Interest and Skill Survey (CISS).**

10. Psychiatric Rating Scales (very high yield)

Depression

- **BDI-II (Beck Depression Inventory):** 21 items, self-report, 0–3 each, total 0–63.
- **HAM-D / HDRS (Hamilton Depression Rating Scale):** clinician-administered, 17 or 21 items.
- **MADRS (Montgomery-Åsberg):** 10 items, clinician.
- **PHQ-9:** 9 items, primary care, maps to DSM.
- **CES-D:** community epidemiology.
- **EPDS (Edinburgh):** postnatal depression.

Anxiety

- **BAI (Beck Anxiety Inventory):** 21 items, self-report.
- **HAM-A:** 14 items, clinician.
- **STAI (Spielberger):** state vs trait anxiety.
- **GAD-7:** primary care.
- **Y-BOCS:** Yale-Brown OCD, 10 items, gold standard for OCD severity.

Psychosis

- **PANSS (Positive and Negative Syndrome Scale):** 30 items, schizophrenia.
- **BPRS (Brief Psychiatric Rating Scale):** 18 items.
- **SAPS / SANS:** Andreasen, separate positive/negative scales.

Mania

- **YMRS (Young Mania Rating Scale):** 11 items.

Cognitive / Dementia

- **MMSE (Mini-Mental State Exam):** Folstein, 30 points. Cutoff ≤ 23/24 suggests impairment.
- **MoCA (Montreal Cognitive Assessment):** 30 points; better for mild cognitive impairment than MMSE.
- **ACE-III (Addenbrooke's):** 100 points.
- **Clock Drawing Test.**

Suicide

- **Beck Scale for Suicide Ideation (BSS).**
- **Columbia Suicide Severity Rating Scale (C-SSRS).**

Trauma / PTSD

- **CAPS-5 (Clinician-Administered PTSD Scale).**
- **PCL-5 (PTSD Checklist).**
- **IES-R (Impact of Event Scale-Revised).**

Substance

- **AUDIT** (alcohol), **DAST** (drugs), **CAGE** (4-item alcohol screen).

General Functioning

- **GAF (Global Assessment of Functioning):** 0–100, DSM-IV axis V (dropped in DSM-5).
- **WHODAS 2.0:** DSM-5 functional measure.

11. Educational Measurement

Achievement vs Aptitude vs Intelligence

- **Achievement:** what you've learned (past). School exams, WIAT.
- **Aptitude:** what you can learn (future). DAT, SAT.
- **Intelligence:** general cognitive capacity. WAIS, SB.

Norm-Referenced vs Criterion-Referenced

- **Norm-referenced:** compare student to peer group. Useful for ranking, selection.
- **Criterion-referenced:** compare student to an absolute standard. Useful for mastery, licensing.

Formative vs Summative

- **Formative:** during learning, to guide instruction. Low stakes.
- **Summative:** end of unit/course. High stakes.

12. Top MCQ Traps (Memorize These)

1. **Reliability ≤ 1; validity ≤ √reliability.** Unreliable tests cannot be valid.
2. **p = 0.50 is ideal item difficulty** (for free-response). For 4-option MCQ, adjust for guessing → ~0.625.
3. **Negative discrimination (D < 0) = miskeyed item.** Drop immediately.

4. **Cronbach's α is not a measure of unidimensionality.**
5. **Spearman-Brown adjusts split-half** because halving a test artificially lowers reliability.
6. **Face validity ≠ content validity.** Face = looks like; content = expert judgment of domain coverage.
7. **Predictive > concurrent** when forecasting is the goal; concurrent when gold standard exists now.
8. **MMPI-2 has 10 clinical + 3+ validity scales.** Don't confuse with the 16 of 16PF.
9. **Rorschach = 10 inkblots** (5 color, 5 B/W). Not 20. Not 12.
10. **TAT = 31 cards total**, 20 used per admin based on age/sex of subject.
11. **Big Five ≠ 16PF.** Big Five (OCEAN) by Costa-McCrae; 16PF by Cattell.
12. **Bhatia Battery has 5 subtests.** Easy to confuse with WAIS-IV's 4 indices.
13. **IQ SD = 15 for Wechsler**, **16 for old Stanford-Binet**. Question often specifies; read carefully.
14. **Likert items are summed; Thurstone items are scale-valued by judges.** Likert is easier.
15. **MMSE max = 30**, ≤ 23/24 = cognitive impairment. MoCA max = 30, ≤ 25 = MCI.
16. **Y-BOCS = OCD**, not anxiety generally. Common distractor.
17. **PANSS = schizophrenia** (both positive + negative). BPRS is briefer.
18. **Stanine mean = 5, SD = 2.** Not 5 and 1. Not 10 and 2.
19. **Kuder-Richardson (KR-20) = Cronbach's α for dichotomous items.** Same idea, different formula.
20. **Inter-rater reliability for categorical = Cohen's κ**, for continuous = ICC.

13. Key Contributors (rote)

- **Alfred Binet:** first practical intelligence test (1905, with Simon).
- **Lewis Terman:** Stanford-Binet; introduced IQ ratio for US use.
- **David Wechsler:** adult intelligence scale; deviation IQ.
- **Raymond Cattell:** 16PF; fluid vs crystallized intelligence.
- **Charles Spearman:** g factor; split-half reliability.
- **Louis Thurstone:** Primary Mental Abilities; equal-appearing intervals.
- **Rensis Likert:** summated rating scale.
- **Hermann Rorschach:** inkblot test.
- **Henry Murray:** TAT; needs-press theory.
- **Starke Hathaway & Charnley McKinley:** MMPI.
- **Paul Costa & Robert McCrae:** NEO-PI-R; Big Five.
- **John Holland:** RIASEC vocational theory.
- **Robert Hogan:** Hogan Personality Inventory.
- **Lee Cronbach:** Cronbach's α (1951); construct validity paper with Meehl (1955).
- **Samuel Messick:** unified validity framework (1989).
- **C.M. Bhatia:** Bhatia Battery (Indian).

Study approach: memorize Sections 3, 4, 10, and 12 cold. Sections 5, 6 need light review. Sections 7, 9, 11, 13 are straight rote. Drill MCQs daily (set-02 incoming).

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Part 3: Diagnostic Review

Baseline Diagnostic 2026-04-22

Baseline Diagnostic, 7 Sections x 5 MCQs

Sent to Sihan on 2026-04-22 at his request (“Test me on each section and assess my current level of knowledge”). Purpose: establish section-wise baseline so the study plan can be rebuilt against actual gaps rather than self-reported weak areas.

Answer Key

Section	Q1	Q2	Q3	Q4	Q5
1. Biopsychology	**c**	**b**	**c**	**d**	**b**
2. Developmental Psych	**a**	**d**	**b**	**b**	**b**
3. Abnormal Psych	**b**	**b**	**c**	**b**	**c**
4. Psych Assessment	**b**	**a**	**b**	**d**	**c**
5. Psychotherapy	**b**	**b**	**c**	**b**	**b**
6. Biostats & Research	**b**	**c**	**b**	**c**	**c**
7. Language / English	**b**	**b**	**b**	**b**	**b**

Scoring

- +4 right / -1 wrong / 0 blank
- Max per section: 20
- Max overall: 140

Section 1: Biopsychology

Q1. The most abundant glial cell in the CNS is: a) Microglia b) Oligodendrocytes c) **Astrocytes** ✓ d) Ependymal cells

Q2. Broca area is located in: a) Superior temporal gyrus of the left hemisphere b) **Inferior frontal gyrus of the left hemisphere** ✓ c) Postcentral gyrus of the parietal lobe d) Cingulate gyrus

Q3. Muscle atonia during sleep is characteristic of: a) Stage 2 NREM b) Stage 3 NREM (delta) c) **REM** ✓ d) Stage 1 NREM

Q4. Cortisol is released by the: a) Anterior pituitary b) Posterior pituitary c) Adrenal medulla d) **Adrenal cortex** ✓

Q5. The dopaminergic neurons of the substantia nigra primarily project to: a) Hippocampus b) **Striatum (caudate and putamen)** ✓ (nigrostriatal pathway — lost in Parkinson's) c) Prefrontal cortex d) Cerebellum

Section 2: Developmental Psychology

Q1. Piaget sensorimotor stage covers: a) **0 to 2 years** ✓ b) 2 to 7 years (preoperational) c) 7 to 11 years (concrete operational) d) 11+ years (formal operational)

Q2. Object permanence emerges in Piaget: a) Concrete operational b) Preoperational c) Formal operational d) **Sensorimotor** ✓ (around 8-12 months)

Q3. Erikson stage for adolescence is: a) Trust vs Mistrust (0-1) b) **Identity vs Role Confusion** ✓ (12-18) c) Intimacy vs Isolation (young adult) d) Generativity vs Stagnation (middle adult)

Q4. Ainsworth Strange Situation originally identified how many attachment classifications? a) 2 b) **3** ✓ (secure, anxious-avoidant, anxious-resistant; disorganized added by Main & Solomon) c) 4 d) 5

Q5. Vygotsky Zone of Proximal Development is: a) Upper limit of cognitive ability at a given age b) **Gap between what a child can do alone vs with help** ✓ c) Concrete operational range d) Critical period for language

Section 3: Abnormal Psychology

Q1. DSM-5 requires Major Depressive Disorder symptoms to last at least: a) 1 week b) **2 weeks** ✓ c) 1 month d) 6 months

Q2. Schneiderian first-rank symptoms are most characteristic of: a) Bipolar disorder b) **Schizophrenia** ✓ (Kurt Schneider 1959; thought broadcast, etc.) c) OCD d) PTSD

Q3. DSM-5 GAD requires excessive worry for at least: a) 1 month b) 3 months c) **6 months** ✓ d) 1 year

Q4. Perfectionism, preoccupation with rules, inflexibility, no obsessions/compulsions: a) OCD (needs obsessions + compulsions) b) **OCPD** ✓ (personality disorder, ego-syntonic) c) Schizotypal PD d) Narcissistic PD

Q5. NEGATIVE symptom of schizophrenia: a) Hallucinations (positive) b) Delusions (positive) c) **Avolition** ✓ (also: alogia, anhedonia, flat affect, asociality) d) Disorganized speech (positive)

Section 4: Psychological Assessment

Q1. SEM when r_xx = 0.81, SD = 20: a) 3.8 b) **8.7** ✓ (SEM = SD·√(1−r) = 20·√0.19 ≈ 8.72) c) 18.0 d) 20.0

Q2. GRE → grad GPA correlation of 0.30: a) **Predictive validity** ✓ (future criterion) b) Concurrent validity c) Content validity d) Face validity

Q3. MMPI-2 clinical scales: a) 8 b) **10** ✓ (Hs, D, Hy, Pd, Mf, Pa, Pt, Sc, Ma, Si) c) 13 d) 16 (16PF has 16 factors — common distractor)

Q4. TAT total cards: a) 10 (Rorschach) b) 15 c) 20 (per administration) d) **31** ✓ (20 shown per admin, selected by age/sex)

Q5. Reliability for dichotomously-scored one-administration test: a) Test-retest (needs 2 sittings) b) Alternate forms (needs 2 forms) c) **KR-20** ✓ (α for 0/1 items) d) Cohen kappa (inter-rater categorical)

Section 5: Psychotherapy & Counselling

Q1. Systematic desensitization developer: a) B.F. Skinner b) **Joseph Wolpe** ✓ (1958) c) Albert Ellis (REBT) d) Aaron Beck (CBT)

Q2. REBT ABC model, C stands for: a) Cognition b) **Consequence** ✓ (Activating event → Belief → Consequence) c) Condition d) Context

Q3. Unconditional positive regard: a) Gestalt therapy b) Behavior therapy c) **Person-centered therapy (Rogers)** ✓ d) Psychoanalysis

Q4. Hot seat and present-moment awareness: a) CBT b) **Gestalt therapy** ✓ (Fritz Perls) c) Logotherapy (Frankl) d) Reality therapy (Glasser)

Q5. Wolpe reciprocal inhibition pairs: a) Operant reward with behavior b) **Relaxation with anxiety-provoking stimuli** ✓ c) Free association with dreams d) Cognitive restructuring with beliefs

Section 6: Biostats & Research Methods

Q1. Type I error: a) Accepting H0 when false (Type II / β) b) **Rejecting H0 when true** ✓ (α / false positive) c) Increasing power (1 − β) d) H0 being true

Q2. Compare means of 3+ independent groups: a) Independent samples t-test (only 2) b) Paired t-test (dependent samples, 2) c) **One-way ANOVA** ✓ d) Chi-square (categorical)

Q3. 2x2 contingency table of frequencies: a) Pearson r (continuous) b) **Chi-square** ✓ (or Fisher exact if cells < 5) c) ANOVA d) t-test

Q4. Pearson r = −0.85: a) Weak negative b) Strong positive c) **Strong negative** ✓ (|r| > 0.7 considered strong) d) No relationship

Q5. Cronbach α = 0.62: a) Excellent b) Acceptable for individual decisions c) **Poor, scale needs revision** ✓ (< 0.70 unacceptable) d) Perfect

Section 7: Language Skills / English

Q1. Synonym of ubiquitous: a) rare (opposite) b) **widespread** ✓ c) ambiguous d) underground

Q2. Antonym of verbose: a) loud b) **concise** ✓ c) articulate (similar) d) eloquent (similar)

Q3. Part of speech of CAREFULLY in "She CAREFULLY reviewed": a) Adjective b) **Adverb** ✓ (modifies verb) c) Preposition d) Conjunction

Q4. Correctly punctuated: a) The patient complained of, headaches, fatigue and insomnia. b) **The patient complained of headaches, fatigue, and insomnia.** ✓ c) The patient complained of: headaches, fatigue, and insomnia. d) The patient complained, of headaches fatigue and insomnia.

Q5. Exacerbate most nearly means: a) Alleviate (opposite) b) **Worsen** ✓ c) Conceal d) Imply

Projected Interpretation Thresholds

Section score	Interpretation
17-20	Strong, maintenance mode
13-16	Moderate, targeted review
9-12	Weak, prioritize in crunch
0-8	Critical, re-foundation needed
Negative (penalties dominated)	Blanks preferable in Phase 2

Final answers + scoring go here after Sihan completes each section.

Results Log

Section 1: Biopsychology — 2/5 (score 5/20)

Sihan answers @ 2026-04-22 00:08: 1c 2b 3b 4c 5a

Q	Key	His	Correct
1	c	c	✓
2	b	b	✓
3	c	b	✗ (sleep-stage trap: picked delta, correct is REM; SAME error as set-01 Q7)
4	d	c	

Q	Key	His	Correct
			✗ (adrenal medulla vs cortex confusion)
5	b	a	✗ (hippocampus vs striatum — didn't know nigrostriatal pathway)

Read: Biopsych is weaker on fresh items than set-01 9/10 suggested. The 9/10 likely had some priming effect from earlier content. Sleep architecture and neuroendocrine pathways are the real gaps. 2-3 dedicated sessions warranted; don't deprioritize biopsych.

Section 2: Developmental Psychology — 5/5 (score 20/20)

Sihan answers @ 2026-04-22 00:10: 1a 2d 3b 4b 5b

All 5 correct.

Read: Dev Psych is a strength, not a concern. Maintenance mode only — light review before exam. Frees hours for weaker sections.

Section 3: Abnormal Psychology — 5/5 (score 20/20)

Sihan answers @ 2026-04-22 00:14: 1b 2b 3c 4b 5c

All 5 correct. DSM durations, Schneider first-rank, OCD vs OCPD, pos/neg symptoms all solid.

Read: Strength section. Maintenance mode.

Section 4: Psychological Assessment — 1/5 (score 0/20)

Sihan answers @ 2026-04-22 00:24: 1b 2b 3a 4a 5d

Q	Key	His	Correct
1	b	b	✓ (SEM math correct)
2	a	b	✗ (predictive vs concurrent confusion)
3	b	a	✗ (MMPI-2 = 10 clinical scales, not 8)
4	d	a	✗ (TAT = 31 total cards; 10 = Rorschach. LITERALLY flashcard Q20 from yesterday)
5	c	d	✗ (KR-20 for internal consistency on dichotomous items; kappa is inter-rater)

Read: This is the weakest section in the diagnostic. Important context: it is the section I built a 13-part master cheatsheet + reliability/validity deep-dive + 20-card flashcard deck for yesterday. Content is in his chat history and the flashcard app; he has NOT yet drilled it.

Revised plan implication: Day 1 (today) becomes two parts — (a) Assessment Deck 1 full drill (20 cards), (b) Assessment Deck 2 (Tests and Scales) drill. Re-test Section 4 after drilling to confirm retention.

Section 5: Psychotherapy & Counselling — 4/5 (score 15/20)

Sihan answers @ 2026-04-22 00:29: 1b 2b 3c 4d 5b

Q	Key	His	Correct
1	b	b	✓ (Wolpe)
2	b	b	✓ (Consequence)
3	c	c	✓ (Rogers)
4	b	d	✗ (Hot seat = Gestalt / Perls, not Glasser's Reality therapy)
5	b	b	✓ (reciprocal inhibition)

Read: Strong on therapy schools and major founders. Gap on less common schools (Gestalt, Reality, Logotherapy, ACT, DBT). Maintenance plus a short read on minor schools.

Section 6: Biostats & Research Methods — 5/5 (score 20/20)

Sihan answers @ 2026-04-22 01:12: 1b 2c 3b 4c 5c

All 5 correct.

Read: Strongest theoretical-abstraction section. Given this is only 10 MCQs on the exam (smallest section), maintenance mode. Low priority for drilling.

Section 7: Language / English — 5/5 (score 20/20)

Sihan answers @ 2026-04-22 01:15: 1b 2b 3b 4b 5b

All 5 correct. (Answer key was legitimately all B.)

Read: English is a free section. Maintenance.

Cumulative Baseline (2026-04-22)

Section	Raw	Score	Status	Priority
1. Biopsychology	2/5	5/20	WEAK	HIGH (sleep + neuroendocrine gaps)

Section	Raw	Score	Status	Priority
2. Developmental	5/5	20/20	STRONG	LOW (maintenance)
3. Abnormal	5/5	20/20	STRONG	LOW (maintenance)
4. Assessment	1/5	0/20	CRITICAL	HIGHEST (material built, not yet drilled)
5. Psychotherapy	4/5	15/20	STRONG	LOW (Gestalt vs Reality note)
6. Biostats & Research	5/5	20/20	STRONG	LOW (maintenance)
7. Language	5/5	20/20	STRONG	LOW (maintenance)
Total	**27/35**	**100/140**	**71%**	—

Projected Exam Score (Extrapolation)

Assuming diagnostic accuracy holds at 100-question scale:

Strategy	Right	Wrong	Blank	Score	Note
Attempt all (current habit)	77	23	0	285	Better than Phase 1 (265)
Strategic blanks (low-conf blanked)	77	10	13	298	
Fix Assessment (1→4) + Biopsych (2→4), attempt all	87	13	0	335	**Clears 320+ target**

Revised Plan Highlights

1. **Today (remainder of 1-hr budget):** Assessment Deck 1 + Deck 2 drill, then reliability/validity top-5 traps.
2. **Apr 23-24:** Biopsych sleep + neuroendocrine focused drilling.
3. **Apr 25:** Re-test Assessment + Biopsych with fresh MCQs to confirm retention.
4. **Apr 26+:** proceed to Abnormal / Psychotherapy foundation per original plan (they are already strong; do not over-invest).
5. **Crunch phase unchanged** in structure, but weighting shifts further toward Assessment + Biopsych since the strong sections are already bankable.

Strategic insight: Sihan's self-assessment was directionally wrong. He named Biopsych #1 weak and Assessment #2 weak. Diagnostic says Assessment is CRITICAL (0/20) and Biopsych is weak but fixable (5/20). Everything else is stronger than he thought. Maintenance, not worry, on Dev/Abnormal/Biostats/Language.